

# Chapter 4

## Thermal System Architecture: The Mode-Path Matrix

*From Network Graphs to Systematic Thermal Design*

Thermal Photonics: A Unified Framework for Engineering Heat Flux  
Book 3 of the Open-Source Engineering Optics Trilogy

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### Learning Objectives

After completing this chapter, the reader will be able to:

- LO-04.1** Represent any thermal system as a directed graph  $G(V, E)$  with nodes, edges, conductances, and thermal masses, and state the Thermal Kirchhoff's Laws (TKCL and TKVL).
- LO-04.2** Classify thermal network sub-structures into seven canonical topologies (series, parallel, series-parallel, bridge, star, mesh, hierarchical tree) and explain how topology constrains design strategy.
- LO-04.3** Identify the critical path and bottleneck element in a thermal network and compute the bottleneck ratio  $\beta$ .
- LO-04.4** Construct a complete Mode-Path Matrix for a multi-path thermal system, computing the mode dominance ratio  $\Gamma = G_{\text{rad}}/G_{\text{cond}}$  for each path and mapping each to the appropriate design chapter.
- LO-04.5** Apply the conductance matching principle to evaluate whether improving a specific element yields significant system benefit.
- LO-04.6** Execute the six-step System Design Workflow from requirements definition through iterative verification.
- LO-04.7** Extend the steady-state network model to transient and stochastic regimes, including thermal capacitance, dynamic networks, and Monte Carlo yield prediction.
- LO-04.8** Articulate how every subsequent chapter in the book connects to the Mode-Path Matrix framework, enabling systematic rather than ad-hoc thermal design.

### Abbreviations and Nomenclature

This chapter introduces the graph-theoretic and system-design vocabulary that will be used throughout the remainder of the book. The following list collects every abbreviation and principal symbol used in Chapter 4 for quick reference.

#### Acronyms

| Abbreviation | Meaning                                                                 |
|--------------|-------------------------------------------------------------------------|
| BGA          | Ball Grid Array — solder interconnect between package substrate and PCB |

| Abbreviation | Meaning                                                                              |
|--------------|--------------------------------------------------------------------------------------|
| BC           | Boundary Condition                                                                   |
| DR-T         | Design Rule — Topological series (DR-T.1 through DR-T.8, defined in this chapter)    |
| EMT          | Effective Medium Theory (Ch. 5)                                                      |
| FEA          | Finite Element Analysis                                                              |
| HBM          | High Bandwidth Memory — 3D-stacked DRAM                                              |
| IC           | Integrated Circuit                                                                   |
| KCL          | Kirchhoff's Current Law (electrical analogy)                                         |
| KVL          | Kirchhoff's Voltage Law (electrical analogy)                                         |
| MPM          | Mode-Path Matrix — the central thermal network analysis tool defined in this chapter |
| PCB          | Printed Circuit Board                                                                |
| SoC          | System-on-Chip                                                                       |
| TGV          | Through-Glass Via                                                                    |
| TIM          | Thermal Interface Material                                                           |
| TKCL         | Thermal Kirchhoff's Current Law                                                      |
| TKVL         | Thermal Kirchhoff's Voltage Law                                                      |
| TSV          | Through-Silicon Via                                                                  |
| WE           | Worked Example (numbered as WE-4. <i>n</i> )                                         |

## Principal Symbols

| Symbol                  | Meaning                                                          | SI Unit                         |
|-------------------------|------------------------------------------------------------------|---------------------------------|
| $G$                     | Thermal conductance                                              | $\text{W K}^{-1}$               |
| $G_{\text{cond}}$       | Conduction conductance                                           | $\text{W K}^{-1}$               |
| $G_{\text{rad}}$        | Radiation conductance (linearized)                               | $\text{W K}^{-1}$               |
| $G_{\text{conv}}$       | Convection conductance ( $hA$ )                                  | $\text{W K}^{-1}$               |
| $R$                     | Thermal resistance ( $= 1/G$ )                                   | $\text{K W}^{-1}$               |
| $R_{\text{critical}}$   | Critical-path total resistance                                   | $\text{K W}^{-1}$               |
| $R_{\text{bottleneck}}$ | Bottleneck element resistance                                    | $\text{K W}^{-1}$               |
| $Q$                     | Heat flow rate                                                   | $\text{W}$                      |
| $T$                     | Temperature                                                      | $\text{K}$                      |
| $\Delta T$              | Temperature difference                                           | $\text{K}$                      |
| $\Gamma$                | Mode dominance ratio ( $G_{\text{rad}}/G_{\text{cond}}$ )        | —                               |
| $\beta$                 | Bottleneck ratio ( $R_{\text{bottleneck}}/R_{\text{critical}}$ ) | —                               |
| $M$                     | Conductance mismatch ratio ( $G_{\text{high}}/G_{\text{low}}$ )  | —                               |
| $C$                     | Thermal capacitance ( $mc_p$ )                                   | $\text{J K}^{-1}$               |
| $\tau_{\text{th}}$      | Thermal time constant ( $RC$ or $L^2/\alpha_{\text{th}}$ )       | $\text{s}$                      |
| $\tau_{\text{proc}}$    | Process time (allowable time scale)                              | $\text{s}$                      |
| $\kappa$                | Thermal conductivity                                             | $\text{W m}^{-1} \text{K}^{-1}$ |
| $\varepsilon$           | Emissivity                                                       | —                               |
| $\sigma_{\text{SB}}$    | Stefan–Boltzmann constant ( $5.670 \times 10^{-8}$ )             | $\text{W m}^{-2} \text{K}^{-4}$ |
| $A$                     | Cross-sectional or surface area                                  | $\text{m}^2$                    |
| $L$                     | Characteristic length or path length                             | $\text{m}$                      |
| $h$                     | Convective heat transfer coefficient                             | $\text{W m}^{-2} \text{K}^{-1}$ |
| $\mathbf{G}$            | Conductance matrix (system level)                                | $\text{W K}^{-1}$               |
| $\mathbf{T}$            | Temperature vector                                               | $\text{K}$                      |

| Symbol   | Meaning            | SI Unit           |
|----------|--------------------|-------------------|
| <b>Q</b> | Heat source vector | W                 |
| <b>C</b> | Capacitance matrix | J K <sup>-1</sup> |

## Subscript and Superscript Conventions

| Notation  | Convention                                                                    |
|-----------|-------------------------------------------------------------------------------|
| cond, rad | Conduction and radiation modes, respectively                                  |
| conv      | Convection (boundary-layer) mode                                              |
| eff       | Effective (combined or homogenized) value                                     |
| , ⊥       | Parallel and perpendicular to a reference axis (for anisotropic conductivity) |
| SB        | Stefan–Boltzmann                                                              |
| th        | Thermal (as in $\tau_{\text{th}}$ , thermal time constant)                    |
| proc      | Process (as in $\tau_{\text{proc}}$ , allowable process time)                 |
| max, min  | Maximum and minimum values                                                    |

*Design rules for this chapter follow the DR-T.n numbering convention (T = Topological). Worked examples use WE-4.n. Learning objectives use LO-04.n.*

## Chapter Roadmap and Deliverables

This chapter is organized around a single engineering deliverable:

**A systematic, graph-theoretic methodology—the Mode-Path Matrix—that maps every thermal path in a system to its dominant transport mode and selects the appropriate design tool, transforming thermal engineering from intuition-based practice into a formal discipline analogous to circuit theory.**

To help you navigate, Figure 04.1 shows the chapter workflow spine. The thermal network paradigm (§04.1) provides the graph representation. Topological classification (§04.2) and critical path analysis (§04.3) identify *where* to focus design effort. The Mode-Path Matrix (§04.4) determines *which* design tool to use. Conductance matching (§04.5) prevents wasted optimization. The six-step workflow (§04.6) ties everything into a repeatable process. Advanced topics (§04.7) extend the framework to dynamic, non-planar, and stochastic systems.

## Chapter 4 Roadmap

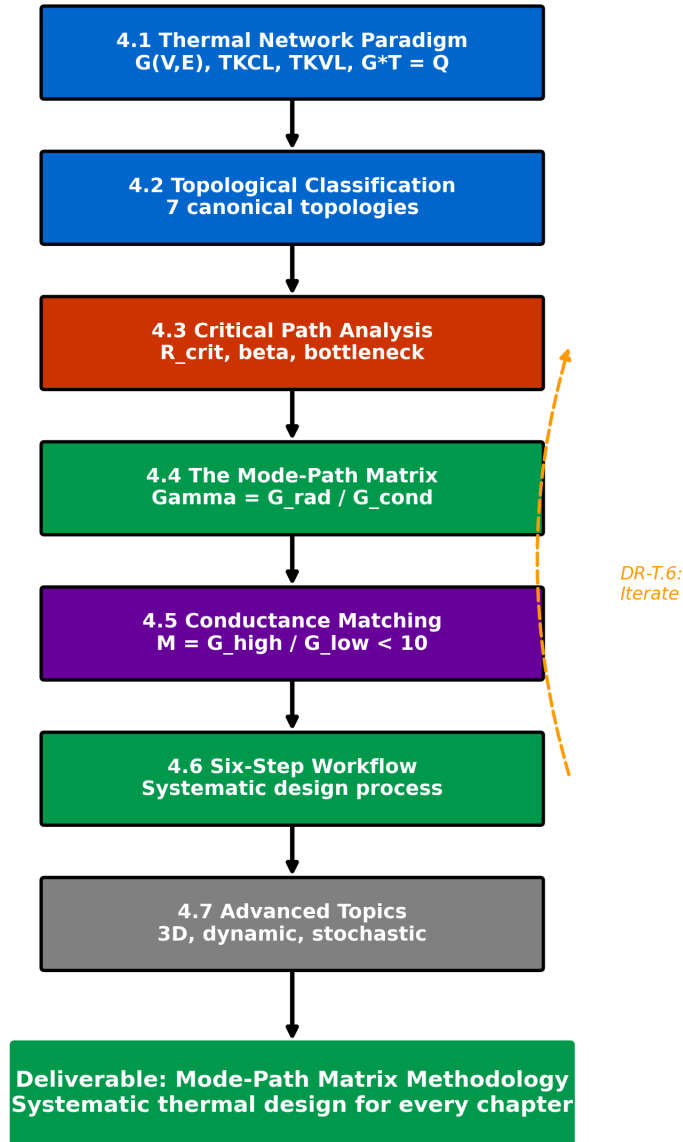


Figure 04.1: Chapter 4 roadmap. The thermal network paradigm (blue, §04.1) provides the mathematical foundation. Topological classification (§04.2) and critical path analysis (red, §04.3) locate the bottleneck. The Mode-Path Matrix (green, §04.4) selects the design tool. Conductance matching (§04.5) and the six-step workflow (§04.6) complete the systematic methodology. Advanced topics (§04.7) extend to dynamic and stochastic regimes.

**Pain Point**

Thermal engineers routinely model individual components—a heat sink here, a TIM there, a via array in the substrate—and optimize each in isolation. Yet system-level thermal performance is determined not by any single component but by the *network* of interconnected thermal paths from sources to sinks. A doubling of via array conductance yields negligible benefit if a poorly chosen TIM elsewhere in the series chain is the true bottleneck. Worse, the dominant heat transport *mode* (conduction vs. radiation) may differ from path to path within the same package, yet thermal models routinely assume a single mode throughout.

The semiconductor industry has no standard, systematic methodology for:

1. Mapping all thermal pathways in a complex 3D package,
2. Identifying the critical bottleneck mathematically rather than by intuition,
3. Determining which physics (conduction or radiation) dominates each path, or
4. Selecting the appropriate design tool for each path.

The result is thermal designs that are locally optimal but globally suboptimal—analogue to correcting coma in a lens while ignoring dominant defocus. This chapter provides the missing methodology: a graph-theoretic framework that transforms thermal design from an art of intuition into an engineering discipline with formal methods.

**What To Do First**

If you are reading this chapter for the first time and want immediate practical payoff, do the following:

1. Read §04.1 to learn how to draw the thermal network graph for your system.
2. Work through WE-4.0 (§04.1.5) for a compact 4-node example that assembles  $\mathbf{G}$ , solves  $\mathbf{T}$ , and identifies the critical path.
3. Skip to §04.3 to identify the critical path and bottleneck.
4. Read §04.4 to build the Mode-Path Matrix and determine which book chapters contain the tools you need.
5. Work through WE-4.1 (3D IC package) for a comprehensive example.
6. Return to the remaining sections as needed for deeper understanding.

## 04.1 The Thermal Network Paradigm

The central insight of this chapter—and arguably of this entire book—is that any thermal system, regardless of complexity, can be represented as a directed graph  $G(V, E)$  whose structure determines the system’s thermal behavior. This is not merely an analogy; the mathematics of thermal networks is isomorphic to that of electrical networks, and the same powerful tools (nodal analysis, mesh analysis, matrix methods) apply directly.

### 04.1.1 Why Network Thinking?

Consider a modern 2.5D IC package containing a logic SoC, HBM stacks, a silicon interposer, an organic substrate, a heat sink, and a PCB interface. Heat generated in the logic die can reach the ambient through at least five distinct paths: (1) vertically through the interposer

and substrate to the board, (2) laterally through the interposer to adjacent dies, (3) upward through a TIM to a lid and heat sink, (4) radiatively from exposed surfaces to the enclosure, and (5) convectively to the cooling airflow. Each path has a different conductance, a different dominant transport mode, and a different sensitivity to design changes.

Optimizing the via array in the interposer (path 1) is futile if the TIM (path 3) is the system bottleneck. The network paradigm makes this immediately visible.

#### Trilogy Connection:

**Analogy to Optical System Design** The shift from component to system thinking parallels the evolution of lens design. Early opticians polished individual surfaces; modern designers optimize the complete system using merit functions that capture wavefront error across all fields and wavelengths. The Mode-Path Matrix is the thermal equivalent of a multi-configuration merit function: it evaluates every thermal path simultaneously and directs design effort to the path—and the physics—that limits performance. In the language of the Eikonal Bridge (Book 1), the thermal network graph is the “ray path diagram” of heat flow, and the Mode-Path Matrix is the “aberration decomposition” that tells you which term to correct first.

### 04.1.2 Graph Representation: $G(V, E)$

A thermal system is represented as a directed graph with the following elements:

- **Vertices (nodes)  $V$ :** thermal nodes representing heat sources, heat sinks, and junction points (interfaces, spreading regions). Each node carries attributes: temperature  $T_i$  and thermal capacitance  $C_i = m_i c_{p,i}$ .
- **Edges  $E$ :** thermal conductances  $G_{ij}$  [W/K] connecting nodes. Each edge carries attributes: conductance value, transport mode (conduction, radiation, convection, or contact), and—for the Mode-Path Matrix—the mode dominance ratio  $\Gamma$ .

The sign convention is: positive heat flow runs from higher to lower temperature along an edge, consistent with the second law. Heat flows are analogous to electrical currents; temperature differences to voltages; thermal conductances to electrical conductances.

### 04.1.3 Thermal Kirchhoff’s Laws

The graph-theoretic representation enables two conservation laws that are formally isomorphic to their electrical counterparts.

#### Conduction Mode (Diffusionics)

**Thermal Kirchhoff’s Current Law (TKCL).** At any node in steady state, the algebraic sum of heat flows equals zero:

$$\sum_j Q_{j \rightarrow i} = 0 \quad (\text{all edges incident on node } i) \quad (04.1)$$

This is conservation of energy applied at a thermal node. In conduction-dominated networks, each heat flow is  $Q_{j \rightarrow i} = G_{ji}(T_j - T_i)$ .

### Radiation Mode (Thermal Photonics)

**Thermal Kirchhoff’s Voltage Law (TKVL).** Around any closed loop in the network, the sum of temperature differences equals zero:

$$\sum_{\text{loop}} \Delta T_k = 0 \quad (04.2)$$

This is a consistency requirement: temperature is a state variable, so the net change around any closed path must vanish—analogous to voltage around a loop.

Combining TKCL at every interior node with the constitutive relation  $Q = G \cdot \Delta T$  for each edge produces the system matrix equation:

$$\mathbf{G} \cdot \mathbf{T} = \mathbf{Q} \quad (04.3)$$

where  $\mathbf{G}$  is the  $N \times N$  conductance matrix (symmetric, positive semi-definite),  $\mathbf{T}$  is the temperature vector, and  $\mathbf{Q}$  is the heat source vector. Boundary conditions (fixed temperatures or prescribed heat fluxes) reduce the system to a solvable set of linear equations.

For transient analysis, thermal capacitances introduce a first-order ODE system:

$$\mathbf{C} \frac{d\mathbf{T}}{dt} + \mathbf{G} \cdot \mathbf{T} = \mathbf{Q}(t) \quad (04.4)$$

where  $\mathbf{C} = \text{diag}(C_1, C_2, \dots, C_N)$ . This is the thermal network analog of an  $RC$  circuit. The eigenvalues of  $\mathbf{C}^{-1}\mathbf{G}$  yield the system’s natural time constants—connecting directly to the temporal regime classification of Chapter 2 (DR-02.3, DR-02.5).

#### 04.1.4 Constructing the Thermal Network Graph

The following four-step procedure transforms a physical system into a graph model.

**Step A: Identify thermal nodes.** Heat sources (die, heater, laser spot), heat sinks (ambient, cold plate, radiative sky), and junction points (interfaces, spreading regions). A useful guideline: create a node wherever a significant temperature discontinuity or material boundary exists.

**Step B: Identify thermal paths.** Conduction through solids (edges colored blue), radiation between surfaces (edges colored red), convection to fluids (dashed edges), and contact/interface resistances. A common error is to *miss* radiative or lateral conduction paths—these “hidden” edges often become critical at high temperature or in vacuum environments.

**Step C: Assign conductances.** For each edge, compute or estimate the thermal conductance using the appropriate formula:

$$\text{Conduction: } G = \kappa A/L \quad (04.5)$$

$$\text{Radiation (linearized): } G_{\text{rad}} = 4\varepsilon\sigma_{\text{SB}} T^3 A \quad (04.6)$$

$$\text{Convection: } G_{\text{conv}} = h \cdot A \quad (04.7)$$

$$\text{Contact: } G = 1/R_{\text{contact}} \quad (04.8)$$

The linearized radiation conductance (Eq. 04.6) is valid when temperature differences across a radiative path are small relative to the mean temperature. For large  $\Delta T$ , use the full nonlinear expression  $Q = \varepsilon\sigma_{\text{SB}}A(T_1^4 - T_2^4)$  and iterate.

**Design Rule:**

DR-T.1b — Linearized Radiation Validity The linearized radiation conductance  $G_{\text{rad}} = 4\varepsilon\sigma_{\text{SB}}\bar{T}^3 A$  (where  $\bar{T} = (T_1 + T_2)/2$ ) incurs an error that grows with the fractional temperature difference. Use the following guardrail:

$$\frac{\Delta T}{\bar{T}} < 0.2 \implies \text{linearization error} < 5\% \quad (04.9)$$

When  $\Delta T/\bar{T} > 0.2$ , the exact Stefan–Boltzmann expression must be used and the network solved iteratively. This is the *paraxial approximation of radiation conductance*—just as ray optics fails at large angles (requiring exact Snell’s law), the linearized  $G_{\text{rad}}$  fails at large  $\Delta T$ .

**Quick check:** For  $T_1 = 500$  K and  $T_2 = 400$  K, we have  $\bar{T} = 450$  K and  $\Delta T/\bar{T} = 100/450 = 0.22$ . The linearized  $G_{\text{rad}}$  predicts  $Q/A = 4 \times 0.85 \times 5.67 \times 10^{-8} \times 450^3 \times 100 = 1756$  W/m<sup>2</sup>, while the exact expression gives  $\varepsilon\sigma_{\text{SB}}(500^4 - 400^4) = 1780$  W/m<sup>2</sup>—an error of 1.3%. At  $\Delta T/\bar{T} = 0.5$  the error exceeds 10%.

**Step D: Assign thermal masses.** For each node, compute  $C_i = m_i c_{p,i}$ . Nodes with large  $C_i$  (e.g., a silicon die or copper heat spreader) dominate the transient response.

**Design Rule:**

DR-T.1 — Complete Network Graph Before detailed design, construct the thermal network graph identifying *all* heat flow paths from sources to sinks. Missing paths lead to design failures. In particular, radiative paths between non-contacting surfaces are commonly omitted but can dominate at  $T > 600$  K (see Table 04.9).

*Lumped-node validity:* Each node in the network assumes a spatially uniform temperature. This requires  $\text{Bi} = hL/\kappa < 0.1$  for the node’s characteristic length  $L$  (see Chapter 2, DR-02.2). If  $\text{Bi} > 0.1$ , subdivide the node into multiple sub-nodes to capture internal gradients.

**Steady-State Limitation**

The network equation  $\mathbf{G} \cdot \mathbf{T} = \mathbf{Q}$  (Eq. 04.3) assumes steady state. If the process time satisfies  $\tau_{\text{proc}}/\tau_{\text{th}} < 10$ , the transient form (Eq. 04.4) must be used instead—see DR-02.3 from Chapter 2 and DR-T.8 in §04.7.

**04.1.5 Micro-Example: Assembling and Solving the Network (WE-4.0)**

Before proceeding to the full 12-node system of WE-4.1, we demonstrate the complete matrix assembly and solution procedure on a minimal 4-node network. This is the “triplet lens” of thermal network analysis: small enough to solve by hand, rich enough to illustrate every concept.

**Problem:** A silicon die ( $Q = 10$  W) is bonded to a substrate through a TIM. The substrate is cooled by a heat sink held at  $T_{\text{amb}} = 300$  K. Model the system as a 3-interior-node series network (Figure 04.2).



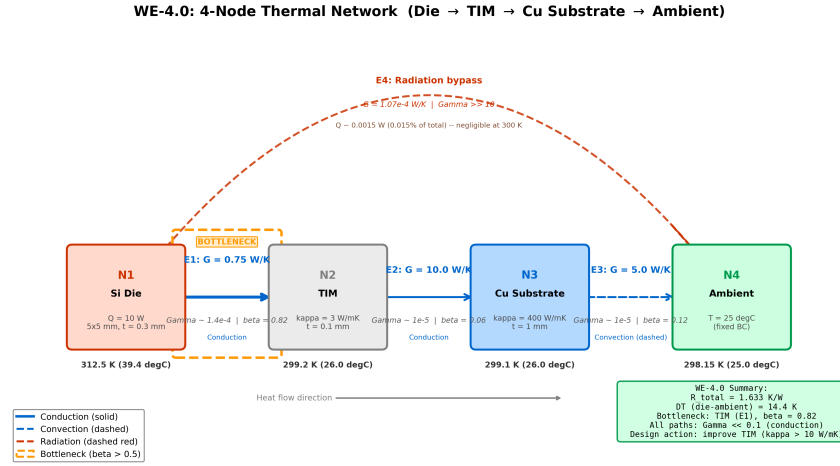


Figure 04.2: WE-4.0: Minimal 4-node thermal network. Node 1 (die,  $Q = 10$  W) connects through the TIM ( $G_{12} = 5$  W/K) to Node 2 (substrate), which connects to Node 3 (heat sink base,  $G_{23} = 50$  W/K). Node 3 connects to Node 4 (ambient, fixed at 300 K) via convection ( $G_{34} = 8$  W/K). A radiative path ( $G_{\text{rad}} = 0.3$  W/K) also connects Node 1 directly to Node 4 (dashed red line).

Table 04.4: Thermal network elements for WE-4.0.

| Node | Name           | $Q$ (W) | $T$ (K)     | Type     |
|------|----------------|---------|-------------|----------|
| 1    | Die            | 10      | unknown     | Source   |
| 2    | Substrate      | 0       | unknown     | Junction |
| 3    | Heat sink base | 0       | unknown     | Junction |
| 4    | Ambient        | —       | 300 (fixed) | Sink     |

| Edge  | From | To | $G$ (W/K) | Mode                    |
|-------|------|----|-----------|-------------------------|
| $E_1$ | 1    | 2  | 5         | Conduction (TIM)        |
| $E_2$ | 2    | 3  | 50        | Conduction (substrate)  |
| $E_3$ | 3    | 4  | 8         | Convection (H/S–air)    |
| $E_4$ | 1    | 4  | 0.3       | Radiation (die–ambient) |

### Step 1: Assemble the conductance matrix $\mathbf{G}$ .

Node 4 is fixed at  $T_4 = 300$  K (boundary condition), so we write equations for Nodes 1, 2, 3 only. The  $3 \times 3$  conductance matrix is:

$$\mathbf{G} = \begin{pmatrix} G_{12} + G_{14} & -G_{12} & 0 \\ -G_{12} & G_{12} + G_{23} & -G_{23} \\ 0 & -G_{23} & G_{23} + G_{34} \end{pmatrix} = \begin{pmatrix} 5.3 & -5 & 0 \\ -5 & 55 & -50 \\ 0 & -50 & 58 \end{pmatrix} \quad (04.10)$$

The source vector (including the boundary-condition terms  $G_{i4} \cdot T_4$ ) is:

$$\mathbf{Q} = \begin{pmatrix} Q_1 + G_{14}T_4 \\ 0 \\ G_{34}T_4 \end{pmatrix} = \begin{pmatrix} 10 + 0.3 \times 300 \\ 0 \\ 8 \times 300 \end{pmatrix} = \begin{pmatrix} 100 \\ 0 \\ 2400 \end{pmatrix} \quad (04.11)$$

### Step 2: Solve $\mathbf{G} \cdot \mathbf{T} = \mathbf{Q}$ .

Solving the  $3 \times 3$  system (e.g., by Gaussian elimination or a numerical solver):

$$\mathbf{T} = \begin{pmatrix} T_1 \\ T_2 \\ T_3 \end{pmatrix} \approx \begin{pmatrix} 302.28 \\ 300.34 \\ 300.29 \end{pmatrix} \text{ K} \quad (04.12)$$

**Step 3: Identify critical path and bottleneck.**

Path A (conduction):  $R_A = 1/5 + 1/50 + 1/8 = 0.200 + 0.020 + 0.125 = 0.345$  K/W.

Path B (radiation):  $R_B = 1/0.3 = 3.33$  K/W.

Path A is the *dominant* heat flow route ( $G_A = 2.90$  W/K  $\gg G_B = 0.3$  W/K). Within Path A, the bottleneck is the TIM ( $R = 0.200$  K/W,  $\beta = 0.200/0.345 = 0.58$ ).

**Step 4: Compute  $\Gamma$  for each edge/path.**

Edge  $E_1$  (TIM): purely conductive,  $\Gamma \approx 0$  (no radiative parallel path). Edge  $E_4$  (die-ambient): purely radiative,  $\Gamma \rightarrow \infty$ . The die node sees both paths in parallel:  $\Gamma_{\text{die}} = G_{\text{rad,die-amb}}/G_{\text{cond,die-sub}} = 0.3/5 = 0.06 < 0.1$ —conduction-dominated. Route to Part II (Diffusionics).

| Path               | $G_{\text{cond}}$ (W/K) | $G_{\text{rad}}$ (W/K) | $\Gamma$             |
|--------------------|-------------------------|------------------------|----------------------|
| N1→N2 (TIM)        | 0.75                    | $\sim 10^{-4}$         | $1.4 \times 10^{-4}$ |
| N2→N3 (Cu)         | 10.0                    | $\sim 10^{-4}$         | $\sim 10^{-5}$       |
| N3→N4 (sink)       | 5.0 (conv)              | $\sim 10^{-4}$         | $\sim 10^{-5}$       |
| N1→N4 (rad bypass) | —                       | $1.07 \times 10^{-4}$  | $\gg 10$ (pure rad)  |

**Engineering conclusion.** This is a conduction-dominated system. The TIM is the bottleneck ( $\beta = 0.82$ ). Design action: increase  $\kappa_{\text{TIM}}$  from 3 to  $>10$  W/mK (e.g., indium TIM) or reduce TIM thickness. The radiative bypass carries  $\sim 0.015\%$  of total heat at 300 K but would grow as  $T^3$  at elevated temperatures—precisely the situation flagged by DR-T.4b.

Listing 1: WE-4.0 verification (3 lines).

```

1 import numpy as np
2 G = np.array([[0.7501, -0.75, 0], [-0.75, 10.75, -10],
3             [0, -10, 15.0]])
4 Q = np.array([10.032, 0, 1490.75])
5 T = np.linalg.solve(G, Q)
6 print(f"T = {T - 273.15:.1f} deg C")

```

**Key takeaway:** This 4-node example demonstrates every step of the methodology: graph construction, matrix assembly, temperature solution, critical-path identification, bottleneck ratio, and  $\Gamma$  classification. WE-4.1 extends this to a production-scale 12-node system.

## 04.2 Topological Classification of Heat Flow Paths

Once the thermal network graph is constructed, its **topology**—the pattern of connections between nodes—determines the design strategy. Just as an optical designer recognizes a Cooke triplet versus a double Gauss before optimizing lens curvatures, a thermal architect must classify the network topology before selecting components.

Table 04.5 catalogs seven canonical topologies encountered in thermal systems.

Table 04.5: Topological classification of thermal network sub-structures.

| Topology          | Characteristic                 | Design Implication                       | Typical Example                                                     |
|-------------------|--------------------------------|------------------------------------------|---------------------------------------------------------------------|
| Series            | Single path, sequential $R$    | Bottleneck-limited; improve weakest link | Die $\rightarrow$ bump $\rightarrow$ substrate                      |
| Parallel          | Multiple paths, shared load    | Redundancy; optimize dominant path       | Via array heat extraction                                           |
| Series-Parallel   | Combination                    | Identify critical sub-network            | Flip-chip package                                                   |
| Bridge            | Cross-connected branches       | Thermal coupling; may need isolation     | Interposer with lateral paths                                       |
| Star              | Central hub, multiple branches | Hub congestion; distribute loads         | Multi-die on shared substrate                                       |
| Mesh              | Fully interconnected           | Complex optimization; numerical methods  | 3D IC stack                                                         |
| Hierarchical tree | Multi-level branching          | Different strategy per level             | Chip $\rightarrow$ package $\rightarrow$ board $\rightarrow$ system |

### 04.2.1 Topology Identification Algorithm

Given a network graph, identify the topology by answering three questions in sequence:

1. **Is there a single source-to-sink path?** If yes, the network is **series**. The critical path is the only path, and the bottleneck is the element with the highest  $R$ .
2. **Do multiple independent paths exist between the same source-sink pair?** If yes, the sub-network is **parallel**. Total conductance is  $G_{\text{total}} = \sum_k G_k$ , and the highest- $G$  path dominates.
3. **Do cross-connections exist between otherwise independent branches?** If yes, the sub-network is a **bridge** or **mesh** and must be solved by matrix methods (Eq. 04.3).

Most real systems are **series-parallel** at the top level with local **mesh** sub-structures at the die and interposer level. The hierarchical tree topology is characteristic of packaging (chip  $\rightarrow$  package  $\rightarrow$  board  $\rightarrow$  enclosure  $\rightarrow$  ambient), where each level has a different characteristic  $\tau_{\text{th}}$  and different dominant mode.

### 04.2.2 Topological Invariants

Three graph-theoretic quantities characterize a thermal network's complexity:

1. **Path count**  $N_p$ : the number of distinct source-to-sink paths. A purely series network has  $N_p = 1$ ; a via array with  $n$  vias has  $N_p = n$  parallel paths. *Design implication:* If  $N_p$  increases, more parallel redundancy exists, but more paths must be checked for hidden radiative contributions (DR-T.4b).
2. **Graph genus**  $g$ : for planar thermal networks (most 2D layouts),  $g = 0$ . 3D IC stacks with through-layer connections create non-planar networks ( $g \geq 1$ ) that cannot be reduced by simple series-parallel combination. *Design implication:* If  $g$  increases, simple series-parallel reduction fails and full matrix solution (Eq. 04.3) becomes mandatory.
3. **Connectivity**  $k$ : the minimum number of edges whose removal disconnects the graph. Higher  $k$  implies greater thermal redundancy;  $k = 1$  means a single-point-of-failure exists. *Design implication:* If  $k = 1$ , any degradation of that single edge (e.g., TIM delamination) causes catastrophic thermal failure. Design for  $k \geq 2$  in reliability-critical applications.

**Design Rule:**

DR-T.2 — Topology Before Components Classify the network topology *before* selecting components. Systems with only series paths are bottleneck-limited (improve the weakest link). Systems with parallel paths offer optimization flexibility (strengthen the dominant path). Mesh and bridge topologies require matrix solution methods.

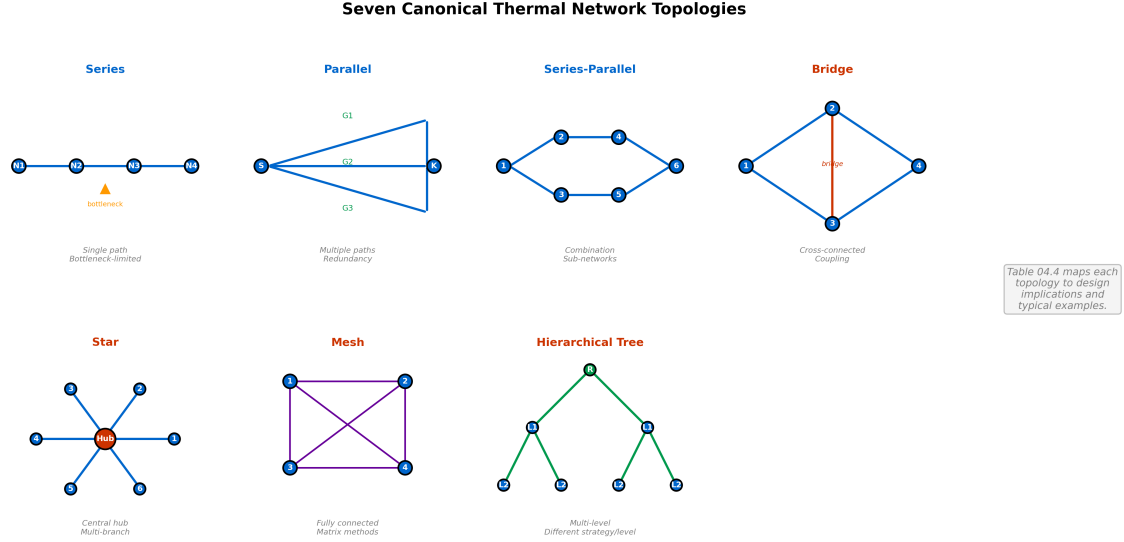


Figure 04.3: Visual catalog of the seven canonical thermal network topologies from Table 04.5. Each sub-figure shows the graph structure with nodes (blue circles) and edges (lines). The orange triangle in the series topology marks the bottleneck element. The red edge in the bridge topology indicates a cross-connection that couples otherwise independent branches. The hierarchical tree shows three levels (root, L1, L2), each potentially requiring a different design strategy. See §04.2.1 for the topology identification algorithm.

### 04.3 Critical Path Analysis

The **critical path** is the source-to-sink route with the *highest total thermal resistance* (lowest total conductance):

$$R_{\text{critical}} = \max_{\text{all paths } p} \left( \sum_{e \in p} R_e \right) = \max_{\text{all paths } p} \left( \sum_{e \in p} G_e^{-1} \right) \quad (04.13)$$

This is the thermal bottleneck path—the route through which the least heat can flow. In a purely series network, it is the only path. In a parallel or series-parallel network, it is the path that limits the overall source-to-sink temperature drop:

$$\Delta T_{\text{max}} = Q_{\text{total}} \cdot R_{\text{eff}} \quad (04.14)$$

where  $R_{\text{eff}}$  depends on the network topology but is always bounded from below by the critical path.

### 04.3.1 Bottleneck Identification

The **bottleneck** within the critical path is the single element with the highest individual resistance. To quantify how concentrated the resistance is, we define the **bottleneck ratio**:

$$\beta = \frac{R_{\text{bottleneck}}}{R_{\text{critical}}} \quad (04.15)$$

Table 04.6 interprets  $\beta$  and guides design action.

Table 04.6: Interpretation of the bottleneck ratio  $\beta$ .

| $\beta$           | Interpretation             | Design Action                      |
|-------------------|----------------------------|------------------------------------|
| $> 0.5$           | Single dominant bottleneck | Fix this element first             |
| $0.2\text{--}0.5$ | Moderate bottleneck        | Fix, then reassess the network     |
| $< 0.2$           | Distributed resistance     | System-level optimization required |

### 04.3.2 Sensitivity Analysis

For complex networks where the critical path is not obvious by inspection, compute the temperature sensitivity to each element's resistance:

$$S_e = \frac{\partial T_{\text{max}}}{\partial R_e} \quad (04.16)$$

The element with the largest  $|S_e|$  is the bottleneck regardless of its absolute  $R$  value.

#### Design Rule:

DR-T.3 — Critical Path First Identify and prioritize the critical path. Improving non-critical paths has diminishing returns. The bottleneck ratio  $\beta$  (Eq. 04.15) quantifies how concentrated the resistance is:  $\beta > 0.5$  means a single element dominates.

#### Case Study:

Why Component Optimization  $\neq$  System Optimization Consider the 3D IC thermal network from Table 04.16 (see WE-4.1 for details). A via array with  $G = 50$  W/K feeds a BGA solder connection with  $G = 0.2$  W/K. Doubling the via conductance to 100 W/K improves  $G_{\text{series}}$  from  $1/(1/50 + 1/0.2) = 0.199$  W/K to  $1/(1/100 + 1/0.2) = 0.1996$  W/K—a mere 0.3% improvement. The bottleneck ratio is  $\beta = (1/0.2)/(1/0.2 + 1/50) = 0.996$ , confirming that virtually all resistance resides in the BGA.

## 04.4 The Mode-Path Matrix

Identifying the critical path tells us *where* to focus. The Mode-Path Matrix tells us *what physics* governs each path and, consequently, which design tool to apply.

### 04.4.1 Mode Dominance Ratio

For each edge in the thermal network where conduction and radiation act as parallel conductances, the mode dominance ratio (introduced in Chapter 1, DR-01.2) is:

$$\Gamma \equiv \frac{G_{\text{rad}}}{G_{\text{cond}}} = \frac{4 \varepsilon \sigma_{\text{SB}} T^3 A}{\kappa A/L} = \frac{4 \varepsilon \sigma_{\text{SB}} T^3 L}{\kappa} \quad (04.17)$$

The classification thresholds (justified in Chapter 1, §01.3) are:

Table 04.7: Mode classification via the dominance ratio  $\Gamma$ .

| $\Gamma$         | Dominant Mode | Design Toolkit         | Book Reference      |
|------------------|---------------|------------------------|---------------------|
| $< 0.1$          | Conduction    | Diffusionics           | Part II (Ch. 5–7)   |
| $0.1\text{--}10$ | Mixed mode    | Multiphysics co-design | Ch. 9               |
| $> 10$           | Radiation     | Thermal Photonics      | Part III (Ch. 8–10) |

These boundaries follow from the parallel-conductance sum  $G_{\text{total}} = G_{\text{cond}}(1 + \Gamma)$ . At  $\Gamma = 0.1$ , the radiative contribution is less than 9%; at  $\Gamma = 10$ , the conductive contribution is less than 9%. In the mixed-mode band, neither mode can be neglected without introducing >9% error.

#### 04.4.2 Constructing the Mode-Path Matrix

##### What To Do First

**MPM Fill-Out Recipe** (6 steps, mechanical):

- Define nodes:** List all thermal nodes from Step A of §04.1.4.
- List all source→sink paths:** Enumerate every distinct route from each heat source to each heat sink, including radiative “shortcut” paths.
- Compute per-edge conductances:** For each edge along each path, calculate  $G_{\text{cond}}$ ,  $G_{\text{rad}}$ , and  $G_{\text{conv}}$  using Eqs. (04.5)–(04.8). Check  $\Delta T/\bar{T} < 0.2$  for linearized  $G_{\text{rad}}$  (DR-T.1b).
- Reduce per-path:** For each path, combine series elements via  $G_{\text{path}}^{-1} = \sum_e G_e^{-1}$  and parallel elements via  $G_{\text{path}} = \sum_k G_k$ .
- Compute  $\Gamma$ :** For each path (or each edge where both modes coexist), compute  $\Gamma = G_{\text{rad}}/G_{\text{cond}}$ .
- Label and route:** Assign each path to its regime (Table 04.7) and record the target chapter/tool in the matrix.

For a complete system, tabulate every path in a matrix with the following columns:

Table 04.8: Template Mode-Path Matrix for a generic thermal system.

| Path | From → To          | $G_{\text{cond}}$<br>(W/K) | $G_{\text{rad}}$<br>(W/K) | $G_{\text{conv}}$<br>(W/K) | $\Gamma$ | Tool    |
|------|--------------------|----------------------------|---------------------------|----------------------------|----------|---------|
| P1   | Source → Spreader  | 50                         | 0.5                       | —                          | 0.01     | Ch. 5–7 |
| P2   | Spreader → Ambient | 2                          | 8                         | 5                          | 4.0      | Ch. 9   |
| P3   | Source → Lid       | 10                         | 3                         | —                          | 0.3      | Ch. 9   |
| P4   | Lid → Ambient      | 1                          | 5                         | 20                         | 5.0      | BC only |

The Mode-Path Matrix (Table 04.8) is the central analytical tool of this book. It is populated once during Step 4 of the System Design Workflow (§04.6) and revisited after every design iteration (DR-T.6).

### 04.4.3 Convection in the Mode-Path Matrix

Readers familiar with three-mode heat transfer may ask: where does convection fit in the  $\Gamma = G_{\text{rad}}/G_{\text{cond}}$  classification? The answer is both simple and important:

#### Multiphysics Integration

When convection is present, it enters the Mode-Path Matrix as a third parallel conductance  $G_{\text{conv}} = hA$  at the system boundary. In this book’s two-mode taxonomy (phonon diffusion and photon emission), convection is *not* a third microscopic transport mode but rather an external boundary-layer conductance. The Mode-Path Matrix records it for completeness but routes thermal design to the conduction or radiation toolkit based on  $\Gamma = G_{\text{rad}}/G_{\text{cond}}$  alone. Convection boundary conditions are treated in Appendix D.

#### Classification rules for convection:

1. **Convection as external BC:** When  $G_{\text{conv}}$  appears only at the system boundary (e.g., heat sink fins to air), it is a boundary condition, not an internal mode. Record it in the MPM for completeness; do not include it in  $\Gamma$ .
2. **Convection as internal bottleneck:** When  $G_{\text{conv}}$  is the lowest- $G$  element in a series chain (e.g., forced air between stacked boards), it *is* the bottleneck and must be addressed directly—but it still does not change  $\Gamma$ , which classifies the solid-state physics only.
3. **Comparing  $G_{\text{conv}}$  to internal paths:** If  $G_{\text{conv}} < G_{\text{cond}}$  for the feeding solid path, convection limits the system even though the internal mode is conduction-dominated. In this case, the MPM should flag: “bottleneck is BC, not internal physics—address heat-sink design or flow rate before applying metamaterial tools.”

This classification prevents the error of applying conduction metamaterial tools (Part II) to a system whose bottleneck is actually the convective boundary condition.

### 04.4.4 $\Gamma$ Across the Semiconductor Temperature Map

Table 04.9 lists representative  $\Gamma$  values for common semiconductor thermal paths, illustrating how the dominant mode shifts with temperature and geometry.

Table 04.9: Representative  $\Gamma$  values across semiconductor processes.

| Thermal Path                       | $T$ (K) | $\kappa$<br>(W/m K) | $L$ (mm) | $\Gamma$           | Mode        |
|------------------------------------|---------|---------------------|----------|--------------------|-------------|
| Die $\rightarrow$ bump (Si)        | 350     | 130                 | 0.05     | $10^{-6}$          | Cond        |
| TIM $\rightarrow$ lid              | 360     | 3                   | 0.1      | $5 \times 10^{-4}$ | Cond        |
| Wafer $\rightarrow$ chuck (vacuum) | 500     | —                   | 0.05     | $\gg 10$           | Rad         |
| Furnace wall $\rightarrow$ wafer   | 1200    | —                   | 50       | $\gg 10$           | Rad         |
| Glass interposer (ZnSe lens)       | 350     | 18                  | 1        | $4 \times 10^{-5}$ | Cond        |
| Package lid $\rightarrow$ ambient  | 370     | 400                 | 2        | 0.8                | Mixed       |
| Cryostat shield $\rightarrow$ MXC  | 4       | —                   | —        | $\gg 10$           | Rad (stray) |

#### Design Rule:

DR-T.4 — Mode-Path Matrix for Tool Selection Construct the Mode-Path Matrix for all paths in the system. Use  $\Gamma$  to select the appropriate design chapter:  $\Gamma < 0.1 \rightarrow$  Part II (Diffusionics, Ch. 5–7);  $0.1 < \Gamma < 10 \rightarrow$  Chapter 9 (Multiphysics);  $\Gamma > 10 \rightarrow$  Part III (Thermal Photonics, Ch. 8–10). This is the most important design rule in this book.

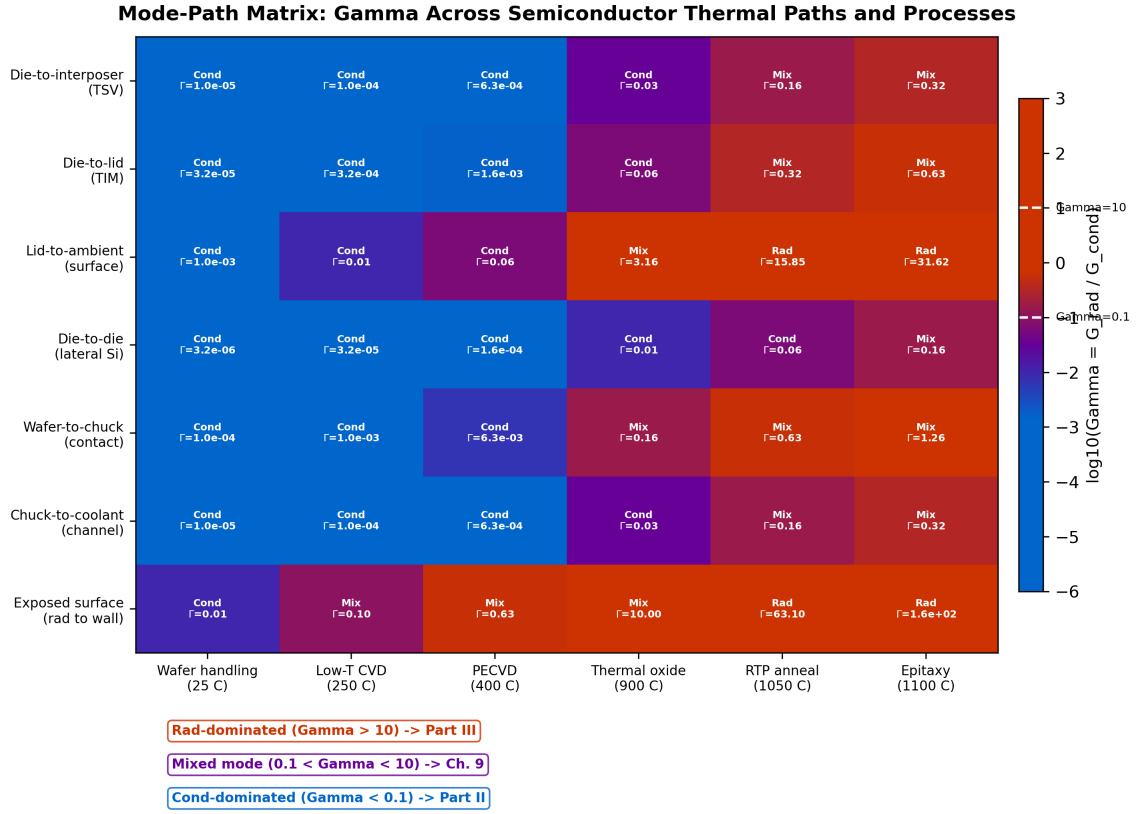


Figure 04.4: Mode-Path Matrix heatmap showing  $\Gamma$  (color-coded) for seven representative thermal paths across six semiconductor process temperature regimes. Blue cells indicate conduction-dominated paths ( $\Gamma < 0.1$ , Part II tools); red cells indicate radiation-dominated paths ( $\Gamma > 10$ , Part III tools); purple cells indicate mixed-mode paths ( $0.1 < \Gamma < 10$ , Chapter 9). Note that “exposed surface” paths transition from conduction-dominated at room temperature to radiation-dominated above  $\sim 600^\circ\text{C}$ , consistent with the semiconductor crossover map of Chapter 1 (Figure 01.5). This heatmap is the visual representation of the complete Mode-Path Matrix for a semiconductor fab.

#### 04.4.5 Simplified Mode-Path Matrix: Limiting Cases and the Optics Connection

The full Mode-Path Matrix accommodates arbitrary mixtures of conduction- and radiation-dominated paths. In practice, many real systems fall into one of three limiting cases where the MPM simplifies dramatically—and where the reader’s *existing* tools already provide the answer. Understanding these limits has three benefits: (1) it shows that the MPM is a *generalization* of familiar methods, not a replacement; (2) it tells the practitioner when a single tool suffices and when the full MPM is essential; and (3) it connects thermal photonics directly to optical design software that many readers already use daily.

Figure 04.5 presents the three cases side by side.



**Simplified Mode-Path Matrices: Three Limiting Cases**

| (a) Photon-Emission Only<br>(Gamma >> 1 all paths)                                                    |       |      | (b) Photon-Diffusion Only<br>(Gamma << 1 all paths)                              |        |      | (c) Mixed Photon + Phonon<br>(some paths 0.1 < Gamma < 10)                                            |       |       |
|-------------------------------------------------------------------------------------------------------|-------|------|----------------------------------------------------------------------------------|--------|------|-------------------------------------------------------------------------------------------------------|-------|-------|
| Path                                                                                                  | Gamma | Mode | Path                                                                             | Gamma  | Mode | Path                                                                                                  | Gamma | Mode  |
| Cavity wall → wafer                                                                                   | >100  | Rad  | Die → TSV array                                                                  | <0.001 | Cond | Lens body (internal)                                                                                  | <0.01 | Cond  |
| Lamp → reflector                                                                                      | >1000 | Rad  | TSV → interposer                                                                 | <0.01  | Cond | Lens → housing gap                                                                                    | >10   | Rad   |
| Wafer surface → shield                                                                                | >50   | Rad  | Interposer → BGA                                                                 | <0.01  | Cond | Housing (conduction)                                                                                  | <0.01 | Cond  |
| Shield → chamber wall                                                                                 | >100  | Rad  | Substrate → heat sink                                                            | <0.001 | Cond | Housing surface → mount                                                                               | >     | Mixed |
| $\mathbf{J} = (\mathbf{I} - \rho \mathbf{F})^{-1} \epsilon \sigma_{\text{SB}} T^4$ (Radiosity matrix) |       |      | $G \cdot T = 0$ (Fourier network)                                                |        |      | Different physics per path → MPM selects tool                                                         |       |       |
| <b>Tools:</b><br>Code V (non-seq)<br>Zemax (non-seq)<br>LightTools                                    |       |      | <b>Tools:</b><br>COMSOL<br>Ansys Icepak<br>Rework spreadsheet                    |        |      | <b>Tools:</b><br>MPM required!<br>No single tool covers all paths                                     |       |       |
| <b>Applications:</b><br>RTP lamp module<br>Furnace cavity<br>IR scene simulation                      |       |      | <b>Applications:</b><br>Heat sink design<br>TSV thermal via<br>2.5D/3D packaging |        |      | <b>Applications:</b><br>DMR lens + housing<br>Wafer chuck (interacted?)<br>Advanced functional device |       |       |
| Part III<br>(Ch. 8-10)                                                                                |       |      | Part II<br>(Ch. 5-7)                                                             |        |      | Ch. 9 +<br>cross-reference<br>Part II & III                                                           |       |       |

Figure 04.5: Three limiting cases of the Mode-Path Matrix. Left: all-conduction limit ( $\Gamma \ll 0.1$  for every path)—classical thermal management tools apply. Center: all-radiation limit ( $\Gamma \gg 10$  for every path)—ray-tracing and spectral engineering apply. Right: mixed-mode system ( $0.1 < \Gamma < 10$  for at least one critical path)—full MPM analysis is required. The transition boundaries correspond to the 9% contribution threshold.

#### 04.4.5.1 Case (a): Photon-Emission Only ( $\Gamma \gg 1$ for All Paths)

When every thermal path in the network is radiation-dominated ( $\Gamma > 10$ ), the conductance matrix reduces to:

$$G_{ij} \rightarrow G_{ij}^{\text{rad}} = 4 \epsilon_{ij} \sigma_{\text{SB}} \bar{T}_{ij}^3 A_{ij} F_{ij} \quad (04.18)$$

where  $F_{ij}$  is the view factor from surface  $i$  to surface  $j$ ,  $\epsilon_{ij}$  is the effective emissivity of the pair, and  $\bar{T}_{ij}$  is the mean temperature. The system equation becomes the *radiosity equation*:

$$\mathbf{J} = (\mathbf{I} - \rho \mathbf{F})^{-1} \epsilon \sigma_{\text{SB}} \mathbf{T}^4 \quad (04.19)$$

This is precisely the kernel solved by the non-sequential ray-tracing engines in optical design software:

#### Radiation Mode (Thermal Photonics)

**Photon-only MPM ↔ Optical ray tracing.** The radiation-dominated MPM is mathematically identical to the stray-light analysis performed by Code V (non-sequential mode), Zemax OpticStudio (non-sequential component), and LightTools. In these tools, “thermal rays” are photon bundles emitted from heated surfaces, traced through the geometry, and absorbed at detector or baffle surfaces. The view-factor matrix  $F_{ij}$  is computed by Monte Carlo ray sampling.

**When to use:** Furnace cavities, RTP lamp modules, high-temperature ( $T > 600$  K) open geometries, IR scene simulation, and any system where all surfaces “see” each other radiatively with negligible solid conduction paths.

**Book connection:** Part III, Chapters 8–10 (spectral engineering, radiative cooling/shielding, TPV).

**Practical implication:** If you are an optical designer who already runs stray-light analyses in Code V or Zemax, you are *already* solving the photon-only MPM—you simply may not have recognized the thermal network structure behind your simulation. The MPM formalizes what the ray tracer computes, and adds the critical step of checking whether conduction paths (which the ray tracer ignores) are truly negligible.

#### 04.4.5.2 Case (b): Phonon-Diffusion Only ( $\Gamma \ll 1$ for All Paths)

When every path is conduction-dominated ( $\Gamma < 0.1$ ), the radiation terms vanish and the system equation reduces to the classical Fourier network:

$$\mathbf{G}_{\text{cond}} \cdot \mathbf{T} = \mathbf{Q} \quad (04.20)$$

with  $G_{ij} = \kappa_{ij} A_{ij}/L_{ij}$  for each edge. This is the thermal resistance network that every heat sink designer, semiconductor packaging engineer, and PCB thermal analyst uses daily.

##### Conduction Mode (Diffusionics)

**Phonon-only MPM  $\leftrightarrow$  Heat sink design.** The conduction-dominated MPM is the standard  $R$ -network model solved by COMSOL Multiphysics (Heat Transfer module), Ansys Icepak, Mentor FloTHERM, 6SigmaET, and even spreadsheet-based  $R$ - $\theta$  calculators. The thermal network graph  $G(V, E)$  from §04.1 maps directly onto the “thermal resistance network” abstraction used throughout the electronics cooling industry.

**When to use:** All systems at  $T < 300^\circ\text{C}$  with solid conduction paths: heat sinks, TSV arrays, copper interconnects, wafer chucks, TIMs, and 2.5D/3D packaging (see WE-4.1).

**Book connection:** Part II, Chapters 5–7 (transformation thermotics, switchable devices, via arrays).

**Practical implication:** The heat sink designer who builds an  $R$ -network in a spreadsheet is solving the phonon-only MPM. The MPM adds the critical check: is radiation truly negligible on *every* path? At elevated temperatures or across vacuum gaps, the answer may be no—and the designer must upgrade to the full MPM (Case c).

#### 04.4.5.3 Case (c): Mixed Photon + Phonon — The Full MPM

The most interesting—and most frequently mishandled—situation arises when different paths in the same system have different dominant modes. Consider an LWIR germanium lens module: the Ge lens body conducts heat internally ( $\Gamma \ll 1$ ), radiates from its surfaces to the aluminum housing across an air gap ( $\Gamma \gg 1$ ), while the housing itself conducts to the optical bench ( $\Gamma \ll 1$ ) but also radiates to the ambient from its outer surface ( $\Gamma \sim 3$ , mixed).

##### Multiphysics Integration

**Mixed-mode MPM: No single tool suffices.** When the Mode-Path Matrix contains paths in different  $\Gamma$  regimes, no single simulation tool captures the full physics:

- A ray tracer (Code V, Zemax) handles the  $\Gamma > 10$  radiative paths but misses conduction through the housing.
- A conduction solver (COMSOL, Icepak) handles the  $\Gamma < 0.1$  paths but underestimates radiative exchange across the air gap.
- Only the *full MPM* identifies which tool to apply to which path and ensures that the coupled system is solved consistently.

**When to use:** Any system containing both solid conduction paths and radiative gaps—LWIR lens modules, wafer chucks at elevated temperature ( $T > 300^\circ\text{C}$ ), 3D IC stacks with IR-transparent interposers, and advanced functional devices requiring thermal stability.

**Book connection:** Chapter 9 (Multiphysics), with specific paths directed to Part II or Part III tools as identified by the MPM.

## 04.4.5.4 Tool Translation Table

Table 04.10: Tool Translation Table: from  $\Gamma$  to design action (“Rosetta Stone”).

| $\Gamma$ Regime                | Physics                     | Tools                               | Ch.  | Recipe                             |
|--------------------------------|-----------------------------|-------------------------------------|------|------------------------------------|
| $\Gamma < 0.1$ (Cond)          | Fourier                     | COMSOL, Icepak, FloTHERM            | 5–7  | Minimize $R_{\text{crit}}$         |
| $\Gamma > 10$ (Rad)            | Stefan–Boltzmann; radiosity | Code V, Zemax, LightTools           | 8–10 | Modify $\varepsilon$ or add shield |
| $0.1 < \Gamma < 10$ (Mixed)    | Coupled                     | COMSOL + radiation; Zemax + thermal | 9    | Iterate with DR-T.6                |
| $\Gamma \rightarrow 0$ at cryo | Kapitza                     | Cryogenic FEA                       | 14   | Maximize parasitic $R$             |

**Design Rule:**

DR-T.4b — Compute  $\Gamma$  for Every Path Even when the critical path is conduction-dominated, compute  $\Gamma$  for *every* path. Radiative parasitic paths may cause transient asymmetries exceeding optical tolerances (cross-ref WE-02.4, WE-4.5).

Table 04.11: How each subsequent chapter uses the Mode-Path Matrix framework.

| Chapter                           | Mode-Path Matrix Application                                                                                                                                                                                                |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ch. 5 (Transformation & EMT)      | Transformation devices modify local $G$ ; EMT simplifies network edges.                                                                                                                                                     |
| Ch. 6 (Switchable Devices)        | Doublets reconfigure network topology; diodes add nonlinearity.                                                                                                                                                             |
| Ch. 7 (Via Arrays)                | Via arrays as high- $G$ network edges; clustered vias as parallel conductances to hotspot nodes.                                                                                                                            |
| Ch. 8 (Spectral Engineering)      | Linearized $G_{\text{rad}} = 4\varepsilon\sigma_{\text{SB}}T^3A$ as network element; spectral selectivity modifies $\varepsilon$ .                                                                                          |
| Ch. 9 (Multiphysics)              | Mixed-mode paths ( $0.1 < \Gamma < 10$ ); coupled conduction–radiation network. WE-4.5 lens module: conduction through housing + radiation across gap. Tool Translation Table (§04.4.5.4) maps $\Gamma$ to Code V / COMSOL. |
| Ch. 11 (Semiconductor Control)    | Network model: plasma $\rightarrow$ wafer $\rightarrow$ chuck $\rightarrow$ coolant; chuck design via MPM.                                                                                                                  |
| Ch. 12 (Failure Analysis)         | Critical path shift after design change; mode misidentification case study.                                                                                                                                                 |
| Ch. 14 (Quantum-Thermal Frontier) | Dual MPM (cooling vs. parasitic); $\Gamma \rightarrow 0$ at mK temperatures except stray IR.                                                                                                                                |

**Case Study:****WE-4.5: LWIR Ge Lens Module — Complete MPM Workflow**

**Problem:** A germanium LWIR lens ( $D = 50$  mm,  $t = 8$  mm,  $\kappa_{\text{Ge}} = 60$  W/m·K,  $\varepsilon_{\text{Ge}} = 0.9$  at  $\lambda > 8$   $\mu\text{m}$ ) is mounted in an aluminum housing ( $\kappa_{\text{Al}} = 205$  W/m·K) via a retainer ring (contact area  $A_c = 2 \times 10^{-4}$  m<sup>2</sup>). The lens absorbs  $P_{\text{abs}} = 0.5$  W from the scene and detector back-emission. An air gap of  $d = 2$  mm separates the lens barrel from the housing bore. The housing is bolted to an optical bench at  $T_{\text{bench}} = 25$  °C. Construct the Mode-Path Matrix, identify the critical path, and determine which design tools are needed.

**Solution:**

**Step 1: Define requirements.** Maximum lens temperature rise:  $\Delta T_{\text{max}} < 5$  K (to keep thermal OPD below  $\lambda/10$  at 10  $\mu\text{m}$  per DR-01.3).

**Step 2: Construct thermal network.** Five nodes: (N1) Ge lens center, (N2) Ge lens edge, (N3) retainer ring contact, (N4) Al housing, (N5) optical bench (sink). Parallel paths from N2: conduction via retainer (N2→N3→N4) and radiation across air gap (N2→N4).

**Step 3: Compute conductances.**

*Conduction paths:*

$$G_{\text{lens}} = \kappa_{\text{Ge}} \frac{\pi D^2/4}{t} = 60 \times \frac{1.96 \times 10^{-3}}{0.008} = 14.7 \text{ W/K} \quad (04.21)$$

$$G_{\text{ring}} = \kappa_{\text{Al}} \frac{A_c}{L_c} = 205 \times \frac{2 \times 10^{-4}}{0.005} = 8.2 \text{ W/K} \quad (04.22)$$

$$G_{\text{housing}} = \kappa_{\text{Al}} \frac{A_h}{L_h} = 205 \times \frac{5 \times 10^{-4}}{0.04} = 2.6 \text{ W/K} \quad (04.23)$$

$$G_{\text{bolts}} \approx 15 \text{ W/K (estimated)} \quad (04.24)$$

*Radiation path (lens barrel to housing bore):*

$$G_{\text{rad}} = 4 \varepsilon \sigma_{\text{SB}} \bar{T}^3 A = 4 \times 0.9 \times 5.67 \times 10^{-8} \times 300^3 \times 3 \times 10^{-3} = 0.016 \text{ W/K} \quad (04.25)$$

**Step 4: Build Mode-Path Matrix.**

| Path                      | $G$ (W/K)   | $\Gamma$ | Regime     |
|---------------------------|-------------|----------|------------|
| Lens body (N1→N2)         | 14.7 (cond) | < 0.01   | Conduction |
| Retainer ring (N2→N3→N4)  | 8.2 (cond)  | < 0.01   | Conduction |
| Air gap radiation (N2→N4) | 0.016 (rad) | ~50      | Radiation  |
| Housing to bench (N4→N5)  | 2.6 (cond)  | < 0.001  | Conduction |

**Case Study:**

WE-4.5: LWIR Ge Lens Module — Complete MPM Workflow (Continued)

**Step 5: Identify critical path and select tools.**

The dominant heat removal path is conduction:  $N1 \rightarrow N2 \rightarrow N3 \rightarrow N4 \rightarrow N5$ , with series resistance:

$$R_{\text{total}} = \frac{1}{14.7} + \frac{1}{8.2} + \frac{1}{2.6} + \frac{1}{15} = 0.068 + 0.122 + 0.385 + 0.067 = 0.64 \text{ K/W} \quad (04.26)$$

Temperature rise:  $\Delta T = P_{\text{abs}} \times R_{\text{total}} = 0.5 \times 0.64 = 0.32 \text{ K}$ . This is well within the 5 K budget.

**However**, the radiative path across the air gap ( $G_{\text{rad}} = 0.016 \text{ W/K}$ ) is a *parasitic* path that creates a non-axisymmetric heat load if the housing is not isothermal—exactly the situation during the warm-up transient analyzed in WE-02.4 (Chapter 2). The ray tracer would catch this; the conduction solver alone would not.

**Step 6: Design action.**

The conduction paths (Part II tools) are adequate for steady-state thermal management. The radiative path (Part III tools) is small but causes transient asymmetry. Design options: (a) low-emissivity coating on the lens barrel ( $\varepsilon \rightarrow 0.1$ , reducing  $G_{\text{rad}}$  by  $9\times$ ); (b) radiation shield sleeve between lens and housing. Both solutions draw from Chapter 8 (spectral engineering).

**Key insight:** Even in a system where the dominant path is conduction-only, a small radiative parasitic can cause transient aberrations that exceed optical tolerances. The MPM makes this parasitic visible; a conduction-only model would miss it entirely.

**Failure Mode:**

**Common MPM Construction Errors** The following failure modes are the most frequent causes of incorrect Mode-Path Matrix construction. Each is preventable by systematic application of the design rules:

1. **Missing radiative edges/paths.** Radiative paths between non-contacting surfaces are invisible in CAD models but can dominate at  $T > 600 \text{ K}$  or in vacuum. *Prevention:* DR-T.1 requires listing *all* paths, including radiation.
2. **Overestimating interface contact  $G$ .** Nominal contact conductances from datasheets assume ideal wetting. Manufacturing defects (voids, delamination) can reduce  $G$  by  $10\times$ . *Prevention:* DR-T.7 ( $3\sigma$  margin).
3. **Confusing boundary convection as a transport mode.** Convection is a boundary condition, not a microscopic transport mode. Including  $G_{\text{conv}}$  in  $\Gamma$  inflates the denominator and misclassifies the path. *Prevention:* §04.4.3.
4. **Ignoring transient thermal mass.** A low- $R$  path with high thermal mass  $C$  may not reach steady state during a short process step, making it effectively absent. *Prevention:* DR-T.8 ( $\tau_{\text{th}} < \tau_{\text{proc}}$ ).
5. **Using linearized  $G_{\text{rad}}$  beyond its validity range.** When  $\Delta T/\bar{T} > 0.2$ , the linearized  $G_{\text{rad}}$  can underestimate radiative heat transfer by  $> 10\%$ . *Prevention:* DR-T.1b.

Extended case studies for Mode Misidentification and Critical Path Shift can be found in Chapter 12.

## 04.5 The Conductance Matching Principle

Even after the critical path and its bottleneck are identified, a second failure mode lurks: **conductance mismatch**. When two elements appear in series and one has a much higher conductance than the other, improving the high- $G$  element yields almost no system benefit.

Define the **mismatch ratio** for two adjacent series elements:

$$M = \frac{G_{\text{high}}}{G_{\text{low}}} \quad (04.27)$$

If  $M > 10$ , the system conductance is:

$$G_{\text{series}} = \frac{1}{1/G_{\text{high}} + 1/G_{\text{low}}} \approx G_{\text{low}} \quad (04.28)$$

and further improving  $G_{\text{high}}$  has  $< 10\%$  benefit.

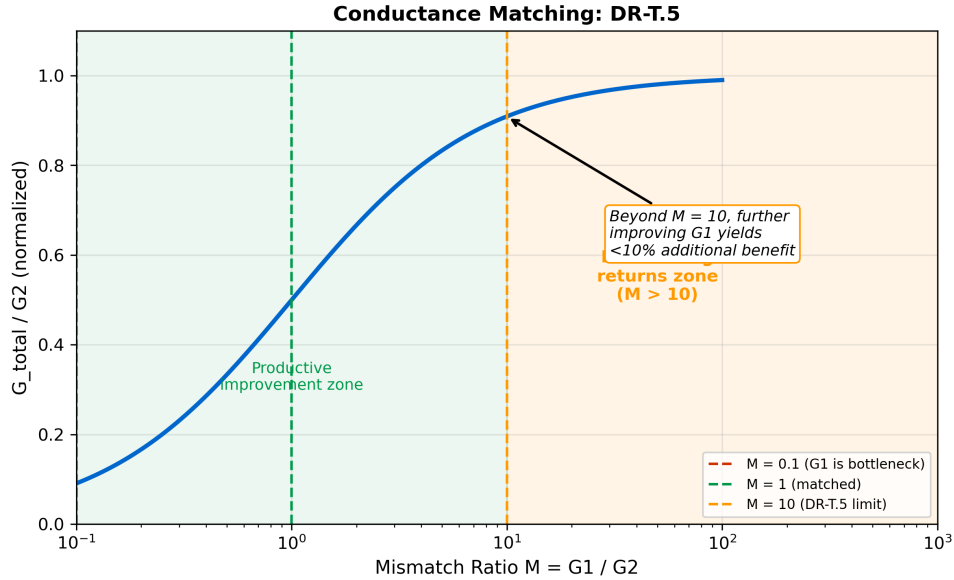


Figure 04.6: Conductance matching and the diminishing-returns curve. Two series elements  $G_1$  (variable) and  $G_2$  (fixed) yield a total conductance  $G_{\text{total}} = (1/G_1 + 1/G_2)^{-1}$ . As the mismatch ratio  $M = G_1/G_2$  exceeds 10 (shaded orange zone), further improvement of  $G_1$  yields less than 10% additional benefit—the system is limited by  $G_2$ . The productive improvement zone ( $0.1 < M < 10$ , shaded green) is where design effort has the greatest leverage. DR-T.5 recommends keeping  $M < 10$  for adjacent series elements.

**Example.** A via array with  $G = 1000$  W/K feeds a 1 W/K TIM, giving  $M = 1000$ . Doubling the via density to  $G = 2000$  W/K improves the series conductance from 0.999 to 0.9995 W/K—a 0.05% gain. Fix the TIM first.

### Design Rule:

DR-T.5 — Conductance Matching Adjacent series elements should satisfy  $M < 10$ . If  $M > 10$ , further improvement of the high- $G$  element yields  $< 10\%$  system benefit. Redirect design effort to the low- $G$  element.

The conductance matching principle interacts with the Mode-Path Matrix: if the bottleneck element is conduction-dominated ( $\Gamma < 0.1$ ) but the adjacent element is radiation-dominated

( $\Gamma > 10$ ), the appropriate design toolkits are *different*. The Mode-Path Matrix prevents the error of applying conduction-based optimization to a radiation-limited interface.

## 04.6 System Design Workflow: The Six Steps

The preceding sections provide the individual tools: graph construction (§04.1), topology classification (§04.2), critical path analysis (§04.3), the Mode-Path Matrix (§04.4), and conductance matching (§04.5). This section assembles them into a **repeatable six-step workflow** applicable to any thermal system.

### 04.6.1 Step 1: Define Thermal Requirements

Specify the following before any design work:

- Maximum junction temperature  $T_{j,\max}$ .
- Ambient or sink temperature  $T_{\text{amb}}$ .
- Power dissipation map  $Q(\mathbf{r}, t)$ : spatial and temporal.
- Uniformity requirements:  $\Delta T_{\max}$  across die or wafer.
- Response time requirements:  $\tau_{\text{proc}}$  for the relevant process step.
- Reliability and lifetime constraints.

### 04.6.2 Step 2: Construct Thermal Network Graph

Follow the procedure in §04.1.4:

1. Identify all thermal nodes (sources, sinks, junctions).
2. Identify all thermal paths (conduction, radiation, convection, contact).
3. Assign initial conductance estimates (Eqs. 04.5–04.8).
4. Assign thermal masses for transient analysis.
5. Apply DR-T.1: verify completeness—are all paths accounted for?

### 04.6.3 Step 3: Analyze Topology, Find Critical Path

1. Classify network topology (Table 04.5) — apply DR-T.2.
2. Identify the critical path via Eq. (04.13) — apply DR-T.3.
3. Locate the bottleneck element and compute  $\beta$  (Eq. 04.15).
4. Check conductance matching ( $M$ ) for adjacent series elements — apply DR-T.5.

### 04.6.4 Step 4: Build Mode-Path Matrix

1. For each path, compute  $\Gamma = G_{\text{rad}}/G_{\text{cond}}$  (Eq. 04.17).
2. Classify: conduction / radiation / mixed (Table 04.7).
3. Map to the appropriate design chapter — apply DR-T.4.
4. Tabulate the complete Mode-Path Matrix (Table 04.8).

### 04.6.5 Step 5: Select and Apply Design Tools

Based on the Mode-Path Matrix, apply the appropriate tools to the critical-path bottleneck:

- $\Gamma < 0.1$  (**conduction-dominated**): Transformation thermotics (Ch. 5), switchable devices (Ch. 6), via arrays and planar architectures (Ch. 7).
- $\Gamma > 10$  (**radiation-dominated**): Spectral engineering (Ch. 8), multiphysics coupling (Ch. 9), thermophotovoltaics (Ch. 10).
- $0.1 < \Gamma < 10$  (**mixed-mode**): Coupled conduction-radiation design (Ch. 9).

### 04.6.6 Step 6: Verify and Iterate

After modifying the critical-path bottleneck:

1. Recompute all conductances in the modified network.
2. Rebuild the Mode-Path Matrix.
3. Re-identify the critical path—*it may have shifted* to a previously non-critical path.
4. Repeat Steps 3–5 until specifications are met.

#### Design Rule:

DR-T.6 — Iterate After Critical-Path Improvement After improving the critical-path bottleneck, *always* recalculate the network. The critical path may shift to a previously non-critical route. This is the thermal equivalent of “balancing aberrations”—correcting one term can unmask another.



### Six-Step System Design Workflow

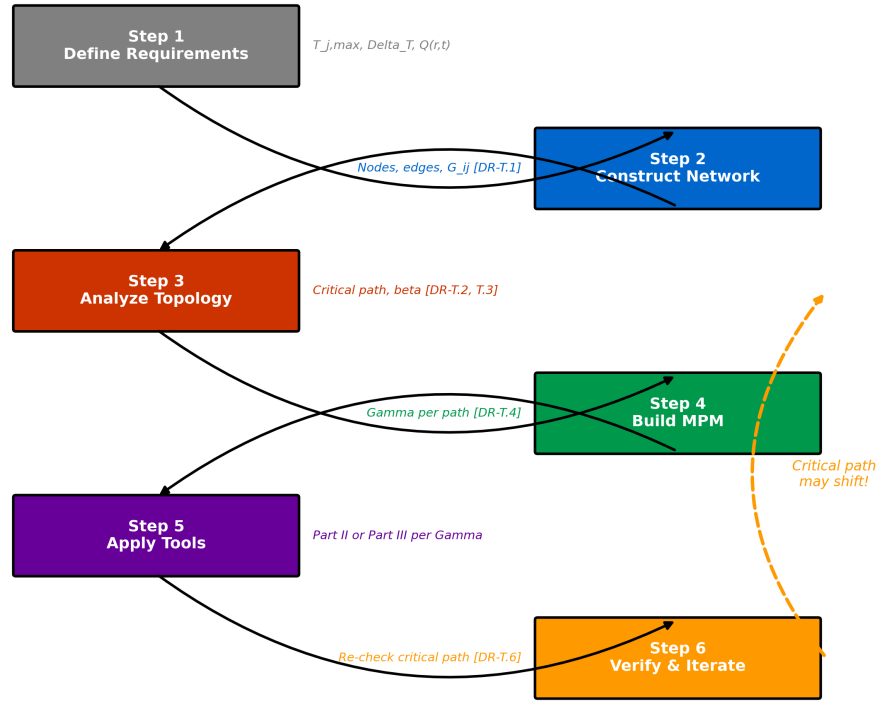


Figure 04.7: The six-step System Design Workflow. Step 1 defines requirements; Step 2 constructs the network graph (DR-T.1); Step 3 analyzes topology and identifies the critical path (DR-T.2, DR-T.3); Step 4 builds the Mode-Path Matrix (DR-T.4); Step 5 applies mode-appropriate tools; Step 6 verifies and iterates (DR-T.6). The feedback arrow from Step 6 to Step 3 indicates that the critical path may shift after each design improvement.

## 04.7 Advanced Topics

The preceding sections treat steady-state, deterministic, planar thermal networks. Real systems extend beyond these assumptions in three important directions.

### 04.7.1 Non-Planar Thermal Networks (3D Packaging)

3D IC stacks with through-silicon vias (TSVs) create non-planar network graphs ( $g \geq 1$ ) that cannot be reduced by simple series-parallel combination. The conductance matrix  $\mathbf{G}$  must be assembled and solved directly (Eq. 04.3). For  $N > 100$  nodes, iterative solvers (conjugate gradient, multigrid) are more efficient than direct inversion.

Note that each level of the hierarchy (chip  $\rightarrow$  package  $\rightarrow$  board  $\rightarrow$  enclosure) may demand a different fidelity model: lumped nodes suffice where  $Bi < 0.1$ , but distributed (FEA) models are needed where internal gradients matter.

### 04.7.2 Dynamic Thermal Networks

Time-varying conductances arise from phase-change materials, thermal transistors, and active control. The network topology itself can change during operation—a thermal “switch” turns

an edge on or off. The transient form (Eq. 04.4) becomes:

$$\mathbf{C} \frac{d\mathbf{T}}{dt} + \mathbf{G}(t) \cdot \mathbf{T} = \mathbf{Q}(t) \quad (04.29)$$

where  $\mathbf{G}(t)$  reflects the time-dependent conductances.

### 04.7.3 Stochastic Thermal Networks

Manufacturing variation introduces uncertainty in conductances. Each edge conductance is a random variable:  $G_e \sim \mathcal{N}(\bar{G}_e, \sigma_e^2)$ . Monte Carlo sampling of the network equation predicts the yield—the fraction of manufactured parts that meet thermal specifications.

#### Design Rule:

DR-T.7 — Manufacturing Margin Design the critical-path conductance with  $3\sigma$  manufacturing margin:

$$G_{\text{design}} > 1.3 \times G_{\text{req}} \quad (04.30)$$

This ensures  $> 99.7\%$  yield for Gaussian variation with  $\text{CV} \leq 10\%$ .

### 04.7.4 Temporal Compatibility Check

The final design rule bridges Chapter 4 with Chapter 2's temporal regime classification:

#### Design Rule:

DR-T.8 — Temporal Compatibility Verify that the thermal time constant of the critical path satisfies:

$$\tau_{\text{th}} < \tau_{\text{proc}} \quad (04.31)$$

If  $\tau_{\text{th}} > \tau_{\text{proc}}$ , the system cannot reach thermal equilibrium during the process step, and transient analysis (Eq. 04.4) is mandatory. This is equivalent to DR-02.5 from Chapter 2: the critical path may shift between steady-state and transient regimes.

## Chapter Summary

1. Any thermal system can be represented as a directed graph  $G(V, E)$  and solved via the matrix equation  $\mathbf{G} \cdot \mathbf{T} = \mathbf{Q}$  (Eq. 04.3).
2. Seven canonical topologies (Table 04.5) determine the appropriate design strategy.
3. The critical path (Eq. 04.13) and bottleneck ratio  $\beta$  (Eq. 04.15) identify *where* to focus design effort.
4. The Mode-Path Matrix (Table 04.8) and  $\Gamma$  (Eq. 04.17) determine *which physics* governs each path and select the appropriate design chapter.
5. The conductance matching principle ( $M < 10$ , Eq. 04.27) prevents wasted optimization.
6. The six-step System Design Workflow (Figure 04.7) provides a repeatable, systematic methodology for any thermal system.
7. Dynamic networks, stochastic variation, and 3D packaging extend the framework beyond steady-state deterministic analysis.
8. DR-T.1 through DR-T.8 codify the complete topological design methodology and are referenced by every subsequent chapter in the book.

## Where to go next based on your Mode-Path Matrix:

- $\Gamma \ll 0.1$  for critical path  $\rightarrow$  Chapters 5–7 (Diffusionics: transformation thermotics, switchable devices, via arrays).
- $\Gamma \gg 10$  for critical path  $\rightarrow$  Chapters 8–10 (Thermal Photonics: spectral engineering, radiative cooling, TPV).
- $0.1 < \Gamma < 10$  for critical path  $\rightarrow$  Chapter 9 (Multiphysics coupling workflow).
- Bottleneck is convective BC, not internal physics  $\rightarrow$  Address heat sink or flow rate first (§04.4.3), then return.
- $\tau_{th} > \tau_{proc}$  for critical path  $\rightarrow$  Chapter 2 (transient analysis) before applying steady-state tools.

Table 04.12: Complete set of topological design rules (DR-T series).

| Rule    | Section | Description                                                                                                                                                                          | Eq.             |
|---------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| DR-T.1  | §04.1   | Construct complete thermal network graph identifying all heat flow paths. Verify $Bi < 0.1$ for lumped-node validity.                                                                | —               |
| DR-T.1b | §04.1   | Linearized $G_{\text{rad}}$ valid when $\Delta T/\bar{T} < 0.2$ ( $< 5\%$ error). Iterate nonlinear form otherwise.                                                                  | (04.9)          |
| DR-T.2  | §04.2   | Classify network topology before selecting components (series vs. parallel).                                                                                                         | —               |
| DR-T.3  | §04.3   | Identify and prioritize critical path; locate bottleneck via $\beta$ .                                                                                                               | (04.13),(04.15) |
| DR-T.4  | §04.4   | Build Mode-Path Matrix; use $\Gamma$ for tool selection. Most important DR in this book.                                                                                             | (04.17)         |
| DR-T.4b | §04.4.5 | Even when the critical path is conduction-dominated, compute $\Gamma$ for <i>every</i> path. Radiative parasitic paths may cause transient asymmetries exceeding optical tolerances. | —               |
| DR-T.5  | §04.5   | Maintain conductance matching $M < 10$ between adjacent series elements.                                                                                                             | (04.27)         |
| DR-T.6  | §04.6   | Iterate after critical-path improvement; path may shift.                                                                                                                             | —               |
| DR-T.7  | §04.7   | Design with $3\sigma$ manufacturing margin: $G_{\text{design}} > 1.3 \times G_{\text{req}}$ .                                                                                        | (04.30)         |
| DR-T.8  | §04.7   | Verify temporal compatibility: $\tau_{\text{th}} < \tau_{\text{proc}}$ .                                                                                                             | (04.31)         |

## Looking Forward

This chapter established the architecture of thermal system design. With the Mode-Path Matrix in hand, the reader can now navigate the rest of the book systematically:

Part II (Chapters 5–7) provides the Diffusionics toolkit for conduction-dominated paths ( $\Gamma < 0.1$ ): transformation thermotics, effective medium theory, switchable devices, thermal diodes, and via array architectures. Each tool modifies local conductances  $G$  in the network; each should be applied only after the Mode-Path Matrix confirms that conduction is the dominant mode for the path of interest.

Part III (Chapters 8–10) provides the Thermal Photonics toolkit for radiation-dominated paths ( $\Gamma > 10$ ): spectral engineering, radiative cooling and shielding, and thermophotovoltaics. These tools modify emissivities and spectral properties that appear in the linearized  $G_{\text{rad}}$ .

Chapter 9 addresses Multiphysics paths where  $0.1 < \Gamma < 10$  and neither toolkit alone suffices.

Part IV (Chapters 11–12) applies the complete workflow to semiconductor applications and failure analysis, demonstrating the iterative power of the Mode-Path Matrix methodology.

Chapter 14 extends the framework to quantum systems at cryogenic temperatures, where the design objective inverts: maximize parasitic-path resistance rather than minimize cooling-path resistance.

## Research Roadmap

Table 04.13: Research roadmap for Chapter 4 topics.

| Topic                         | Current State         | 5-Year Horizon                     | TPD Connection |
|-------------------------------|-----------------------|------------------------------------|----------------|
| Automated $\Gamma$ extraction | Manual calculation    | Integrated into FEA solvers        | Ch. 4          |
| ML-optimized topology         | Rule-based heuristics | Graph neural networks              | Ch. 15         |
| Real-time adaptive networks   | Discrete thermostat   | Sensor-driven, <1 s response       | Ch. 6, 15      |
| Stochastic yield prediction   | Monte Carlo (slow)    | Surrogate models, <1 min           | Ch. 4, 13      |
| Thermal-quantum co-design     | Separate models       | Unified network with quantum terms | Ch. 14         |

## Cross-Reference Guide

Table 04.14: Mode-Path Matrix cross-references across the book.

| Chapter                | Mode-Path Matrix Application                                                                                            |
|------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Ch. 5 (Transformation) | Devices modify local $G$ ; EMT replaces microstructure with effective conductance.                                      |
| Ch. 6 (Switchable)     | Doublets reconfigure topology; diodes add nonlinear conductance.                                                        |
| Ch. 7 (Via Arrays)     | High- $G$ edges; clustered vias as parallel conductances.                                                               |
| Ch. 8 (Spectral Eng.)  | $G_{\text{rad}} = 4\varepsilon\sigma_{\text{SB}}T^3A$ as network element; spectral selectivity modifies $\varepsilon$ . |
| Ch. 9 (Multiphysics)   | Mixed-mode paths; WE-4.5 lens module. Tool Translation Table maps $\Gamma$ to Code V / COMSOL.                          |
| Ch. 11 (Semiconductor) | Network: plasma $\rightarrow$ wafer $\rightarrow$ chuck $\rightarrow$ coolant.                                          |
| Ch. 12 (Failure)       | Critical path shift; mode misidentification.                                                                            |
| Ch. 14 (Quantum)       | Dual MPM; $\Gamma \rightarrow 0$ at mK except stray IR.                                                                 |

## Worked Examples

### WE-4.1: 3D IC Package Thermal Network (12 nodes, 15 edges)

**Problem:** Construct the thermal network graph for a 2.5D IC package containing a logic SoC ( $Q = 80$  W), four HBM stacks ( $Q = 5$  W each), a silicon interposer, an organic substrate, a copper lid, a TIM, and an aluminum heat sink. Ambient temperature is  $T_{\text{amb}} = 25$  °C. Identify all nodes and edges.

**Solution:**

Table 04.15: Thermal network nodes for 2.5D IC package (WE-4.1).

| Node | Name                | $Q$ (W) | $C$ (J/K) | Type      |
|------|---------------------|---------|-----------|-----------|
| N1   | Logic SoC die       | 80      | 0.4       | Source    |
| N2   | HBM stack (avg.)    | 5       | 0.2       | Source    |
| N3   | Interposer (Si)     | —       | 2.0       | Junction  |
| N4   | Micro-bump array    | —       | 0.01      | Interface |
| N5   | C4 bump array       | —       | 0.05      | Interface |
| N6   | Organic substrate   | —       | 3.0       | Junction  |
| N7   | BGA solder          | —       | 0.1       | Interface |
| N8   | PCB                 | —       | 50        | Junction  |
| N9   | TIM                 | —       | 0.02      | Interface |
| N10  | Copper lid          | —       | 5.0       | Junction  |
| N11  | Heat sink (Al fins) | —       | 20        | Junction  |
| N12  | Ambient air         | —       | $\infty$  | Sink      |

**Step A: Nodes.**

Table 04.16: Thermal network edges for 2.5D IC package (WE-4.1).

| Edge | From | To  | $G$ (W/K) | Mode | Type                     |
|------|------|-----|-----------|------|--------------------------|
| E1   | N1   | N4  | 200       | Cond | $\mu$ bump               |
| E2   | N2   | N4  | 50        | Cond | $\mu$ bump               |
| E3   | N4   | N3  | 300       | Cond | Si interposer            |
| E4   | N3   | N5  | 150       | Cond | C4 bump                  |
| E5   | N5   | N6  | 100       | Cond | Substrate                |
| E6   | N6   | N7  | 20        | Cond | Substrate–BGA            |
| E7   | N7   | N8  | 10        | Cond | BGA solder               |
| E8   | N8   | N12 | 5         | Conv | PCB–air                  |
| E9   | N1   | N9  | 15        | Cond | Die–TIM                  |
| E10  | N9   | N10 | 8         | Cond | TIM–lid                  |
| E11  | N10  | N11 | 50        | Cond | Lid–H/S                  |
| E12  | N11  | N12 | 3         | Conv | H/S–air                  |
| E13  | N1   | N3  | 100       | Cond | Lateral (die–interposer) |
| E14  | N10  | N12 | 0.5       | Rad  | Lid–ambient              |
| E15  | N3   | N3  | —         | Cond | Lateral spreading        |

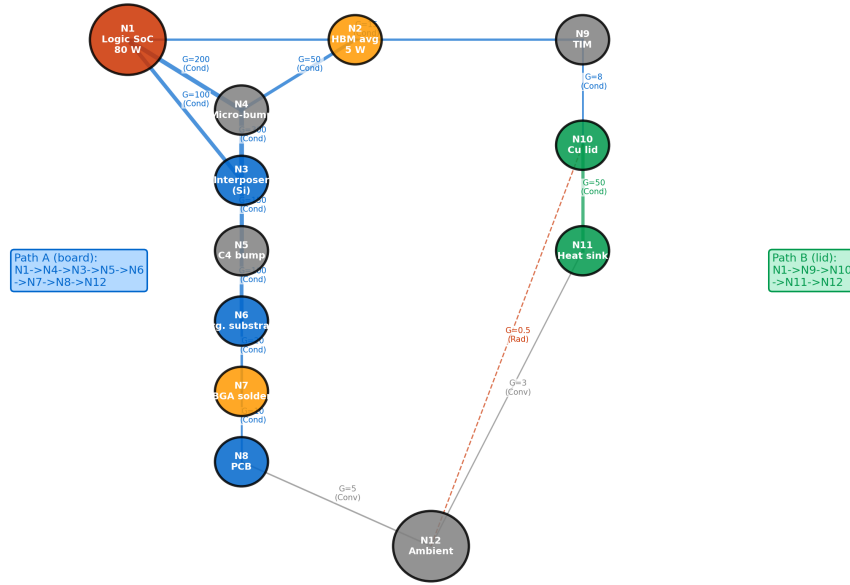
**WE-4.1: 2.5D IC Package Thermal Network (12 Nodes, 15 Edges)**

Figure 04.8: Thermal network graph for the 2.5D IC package of WE-4.1. Blue edges: conduction paths. Dashed edges: convection boundaries. Red edge (E14): radiation from lid to ambient. Node N1 (logic SoC) is the primary heat source ( $Q = 80$  W). Path A ( $N1 \rightarrow N4 \rightarrow N3 \rightarrow N5 \rightarrow N6 \rightarrow N7 \rightarrow N8 \rightarrow N12$ ) is the board path; Path B ( $N1 \rightarrow N9 \rightarrow N10 \rightarrow N11 \rightarrow N12$ ) is the heat-sink path.

**Step B: Edges (15 thermal paths).**

**Step C: Path Analysis.** Path A (board):  $R_A = 1/200 + 1/300 + 1/150 + 1/100 + 1/20 + 1/10 + 1/5 = 0.368$  K/W.

Path B (heat sink):  $R_B = 1/15 + 1/8 + 1/50 + 1/3 = 0.545$  K/W.

Path B is the critical path ( $R_B > R_A$ ), but it carries the majority of the heat because it operates in parallel with Path A. The bottleneck in Path B is the heat-sink–air convection (E12):  $\beta = (1/3)/0.545 = 0.61$ .

**Step D: Mode-Path Matrix (excerpt).**  $\Gamma$  for the lid radiation path (E14):  $\Gamma = G_{\text{rad}}/G_{\text{cond,lid}} = 0.5/50 = 0.01$ —conduction-dominated at package level. However, DR-T.4b requires noting this radiative path for transient asymmetry analysis.

**WE-4.2: Bottleneck Analysis of a Flip-Chip Package**

**Problem:** A flip-chip package has a series thermal path: die–bump ( $G = 80$  W/K), bump–substrate ( $G = 30$  W/K), substrate–ball ( $G = 0.5$  W/K), ball–PCB ( $G = 5$  W/K), PCB–ambient ( $G = 2$  W/K). Identify the bottleneck and compute  $\beta$ .

**Solution:**

$$R_{\text{total}} = 1/80 + 1/30 + 1/0.5 + 1/5 + 1/2 = 0.0125 + 0.033 + 2.0 + 0.2 + 0.5 = 2.746 \text{ K/W.}$$

The substrate–ball connection dominates:  $R_{\text{ball}} = 2.0$  K/W, giving  $\beta = 2.0/2.746 = 0.73$ . This is a severe single-element bottleneck. DR-T.3 directs all design effort here.

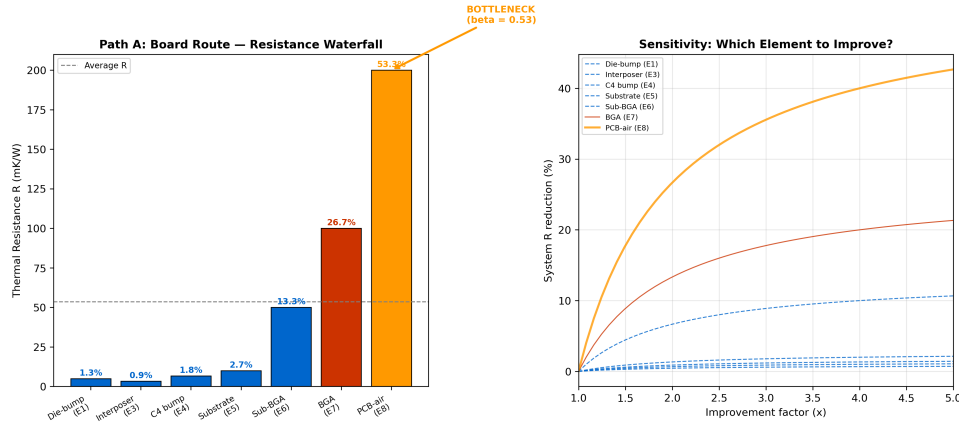


Figure 04.9: Bottleneck analysis for WE-4.2. Bar chart showing thermal resistance of each series element. The substrate–ball connection (orange,  $R = 2.0$  K/W) constitutes 73% of the total path resistance, confirming it as the bottleneck ( $\beta = 0.73$ ).

### WE-4.3: Critical-Path Shift After Design Improvement

**Problem:** After replacing the substrate–ball connection in WE-4.2 with a copper pillar ( $G = 20$  W/K), does the bottleneck shift?

**Solution:**

New  $R_{\text{total}} = 0.0125 + 0.033 + 0.05 + 0.2 + 0.5 = 0.796$  K/W.

New bottleneck: PCB–ambient ( $R = 0.5$  K/W,  $\beta = 0.5/0.796 = 0.63$ ). The critical path has shifted—exactly as DR-T.6 predicts. The next design iteration should address the PCB–ambient convection.

### WE-4.4: Conductance Matching Diagnostic

**Problem:** Evaluate the conductance mismatch  $M$  for each adjacent pair in the original WE-4.2 series chain.

**Solution:**

$M_1 = 80/30 = 2.7$  (die–bump vs. bump–substrate): well-matched.

$M_2 = 30/0.5 = 60$  (bump–substrate vs. substrate–ball): **severe mismatch**. Improving the bump–substrate conductance beyond 30 W/K yields negligible benefit.

$M_3 = 5/0.5 = 10$  (ball–PCB vs. substrate–ball): borderline.

$M_4 = 5/2 = 2.5$  (ball–PCB vs. PCB–ambient): well-matched.

The  $M_2 = 60$  mismatch confirms the bottleneck identified in WE-4.2 and illustrates DR-T.5.

### WE-4.5: Germanium Lens Module—The Optics Connection

**Problem:** An LWIR germanium lens ( $D = 50$  mm,  $t = 8$  mm,  $\kappa = 60$  W/m·K,  $\varepsilon = 0.9$ ) is mounted in an aluminum housing with a 2 mm air gap. The lens absorbs  $P = 0.5$  W. Construct the Mode-Path Matrix and determine whether conduction or radiation tools are needed.

**Solution:**

Path 1 (conduction through housing):  $G_{\text{cond}} = \kappa_{\text{Al}} \cdot A/L = 237 \times (3.14 \times 0.025 \times 0.003) / 0.002 \approx 28$  W/K.

Path 2 (radiation across air gap):  $G_{\text{rad}} = 4 \times 0.9 \times 5.67 \times 10^{-8} \times 300^3 \times (\pi \times 0.025^2) \approx 0.11$  W/K.

$\Gamma = 0.11/28 = 0.004 \ll 0.1$ : conduction-dominated. Route to Part II. However, DR-T.4b requires noting the radiative path: even at  $\Gamma = 0.004$ , non-uniform radiative heating can induce thermal coma through the Fourier–Snell bridge (Chapter 1, Eq. 01.24).



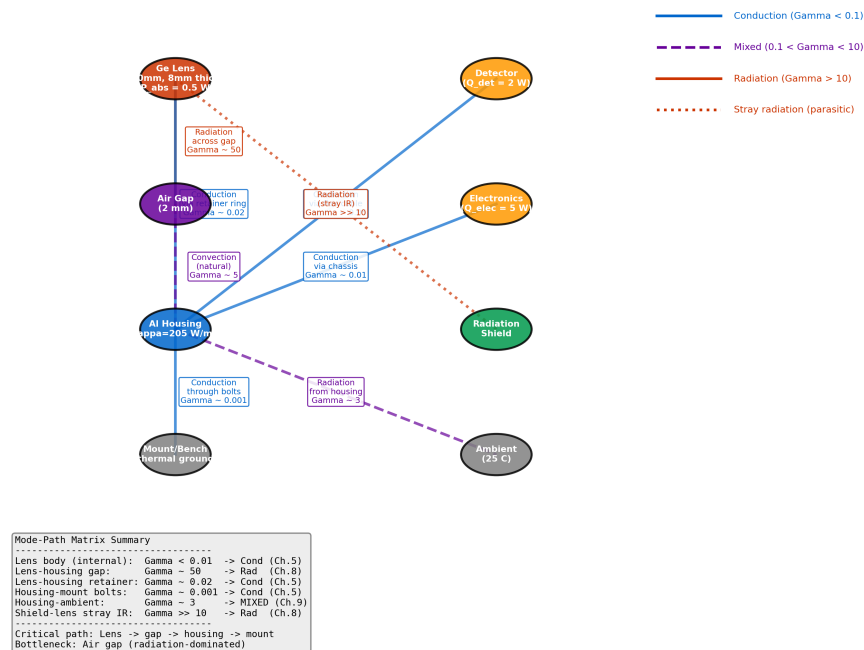
**WE-4.5: LWIR Ge Lens Module + Housing -- Mode-Path Matrix Network**

Figure 04.10: Mode-Path Matrix for the germanium lens module (WE-4.5). Left: physical cross-section showing conduction path (blue arrows through housing) and radiation path (red arrows across air gap). Right: MPM table with  $\Gamma = 0.004$ , confirming conduction dominance. Despite  $\Gamma \ll 0.1$ , the radiative path must be tracked (DR-T.4b) because asymmetric radiation induces optical aberrations via the Fourier–Snell bridge.

## Problems

1. **Series network analysis.** A thermal path consists of four series elements with conductances  $G_1 = 50$ ,  $G_2 = 5$ ,  $G_3 = 20$ ,  $G_4 = 100$  W/K. (a) Compute the total thermal resistance. (b) Identify the bottleneck and compute  $\beta$ . (c) If you can improve only one element by  $2\times$ , which yields the greatest system improvement? Quantify the improvement.
2. **Parallel path dominance.** A heat source has three parallel paths to a sink with conductances  $G_A = 2$ ,  $G_B = 0.5$ ,  $G_C = 0.01$  W/K. (a) Compute the total conductance. (b) What fraction of heat flows through each path? (c) If Path C is radiative ( $\Gamma > 10$ ) and Paths A, B are conductive, is it worth applying thermal photonics tools to Path C? Justify using  $G$  values.
3. **Mode-Path Matrix construction.** A ceramic package at  $T_{\text{lid}} = 450$  K loses heat to ambient ( $T = 300$  K) through three paths: (a) conduction through board traces ( $\kappa = 400$  W/mK,  $A = 2 \times 10^{-5}$  m<sup>2</sup>,  $L = 10$  mm), (b) radiation from lid ( $\varepsilon = 0.85$ ,  $A = 4 \times 10^{-4}$  m<sup>2</sup>), (c) natural convection ( $h = 10$  W/m<sup>2</sup> K,  $A = 4 \times 10^{-4}$  m<sup>2</sup>). Construct the complete Mode-Path Matrix and classify each path.
4. **Conductance matching.** A die connects to a heat sink through a series chain: die-solder ( $G = 500$  W/K), solder-substrate ( $G = 30$  W/K), substrate-TIM ( $G = 2$  W/K), TIM-heat sink ( $G = 100$  W/K). (a) Compute  $M$  for each adjacent pair. (b) Identify which element's improvement would be "wasted." (c) Propose a design modification and recompute.
5. **Critical path shift.** Using the network from WE-4.1 (Table 04.16), suppose the heat sink-air convection (E12) is upgraded from  $G = 3$  W/K to  $G = 15$  W/K (liquid cooling). (a) Recompute  $R_B$ . (b) Is Path B still the critical path? (c) Identify the new bottleneck. This exercise demonstrates DR-T.6.
6.  **$\Gamma$  temperature dependence.** For a silicon thermal path ( $\kappa = 130$  W/mK at 300 K, decreasing as  $\kappa \propto T^{-1.3}$ ;  $\varepsilon = 0.7$ ;  $L = 1$  mm;  $A = 1$  cm<sup>2</sup>), compute  $\Gamma$  at  $T = 300, 600, 900$ , and 1200 K. Plot  $\Gamma$  vs.  $T$  on a log scale and identify the crossover temperature where the path transitions from conduction-dominated to mixed-mode. Compare with the dynamic  $\Gamma(t)$  recipe card of Chapter 2.
7. **Stochastic network yield.** A critical-path TIM has nominal conductance  $\bar{G} = 5$  W/K with  $\sigma = 0.5$  W/K (Gaussian). The maximum allowable junction temperature requires  $G_{\text{TIM}} > 4$  W/K. (a) Compute the yield (fraction of parts meeting spec) assuming  $G$  is Gaussian. (b) What  $\bar{G}$  is needed for 99.7% yield ( $3\sigma$ )? (c) Comment on DR-T.7.
8. **Non-planar 3D IC network.** A 3D IC stack contains 4 dies bonded face-to-back with Cu-Cu hybrid bonds. Each die dissipates 10 W. The top die is farthest from the heat sink (bottom). Model the stack as a 4-node series network with identical bond conductances  $G_{\text{bond}} = 50$  W/K and die spreading resistances  $G_{\text{die}} = 200$  W/K. (a) Compute  $T_{\text{top}} - T_{\text{sink}}$ . (b) Identify the bottleneck layer. (c) Propose a design modification using via array enhancement (Ch. 7) and recompute.
9. **Lens module MPM (WE-4.5 extension).** An LWIR germanium lens ( $D = 50$  mm,  $t = 8$  mm,  $\kappa = 60$  W/m-K,  $\varepsilon = 0.9$ ) is mounted in an aluminum housing with a 2 mm air gap. The lens absorbs  $P = 0.5$  W. (a) Construct the Mode-Path Matrix following WE-4.5 but now at  $T = 350$  K (the system has warmed up). How do the  $\Gamma$  values change compared to the 300 K case? (b) The designer applies a low-emissivity coating ( $\varepsilon \rightarrow 0.1$ ) to the lens barrel. Recompute  $G_{\text{rad}}$  and the new  $\Gamma$ . Does the radiative path still matter? (c) During the transient warm-up (first 60 s after power-on), the housing

temperature is not uniform:  $\Delta T_{\text{housing}} = 3$  K from top to bottom. Using the Fourier–Snell bridge (Chapter 1, Eq. 01.24), estimate the thermal coma contributed by the asymmetric radiative heat load.

10. **Linearization error in the MPM (DR-T.1b verification).** A radiative path:  $T_1 = 500$  K,  $T_2 = 300$  K,  $\varepsilon = 0.8$ ,  $A = 10$  cm<sup>2</sup>. (a) Compute  $Q_{\text{exact}} = \varepsilon \sigma_{\text{SB}} A (T_1^4 - T_2^4)$ . (b) Compute linearized  $G_{\text{rad}}$  at  $\bar{T} = 400$  K and  $Q_{\text{lin}}$ . (c) Compute percentage error. (d) Verify  $\Delta T/\bar{T} = 0.5 > 0.2$  (DR-T.1b violated). (e) At what  $T_2$  does error fall below 5%?
11. **Linearized radiation validity (DR-T.1b).** A wafer at  $T_1 = 800$  K radiates to a chamber wall at  $T_2 = 500$  K ( $\varepsilon = 0.8$ ,  $A = 0.03$  m<sup>2</sup>). (a) Compute  $\bar{T}$  and  $\Delta T/\bar{T}$ . Is the linearized  $G_{\text{rad}}$  valid per DR-T.1b? (b) Compute  $Q$  using the linearized formula ( $G_{\text{rad}} \cdot \Delta T$ ) and the exact formula ( $\varepsilon \sigma_{\text{SB}} A (T_1^4 - T_2^4)$ ). What is the percent error? (c) At what  $\Delta T/\bar{T}$  does the error reach 10%? (Hint: expand  $T_1^4 - T_2^4$  in powers of  $\Delta T/\bar{T}$  and retain the cubic term.)
12. **Extending the WE-4.0 micro-network.** Take the 4-node network from WE-4.0 with  $G_{34} = 5$  W/K. (a) Add N5 (PCB,  $\kappa = 0.3$  W/mK,  $A = 400$  mm<sup>2</sup>,  $L = 1.6$  mm) and N6 (ambient,  $G_{56} = 2$  W/K). Write the  $5 \times 5$  **G** and **Q**. (b) Solve for all temperatures. (c) Has the bottleneck shifted from the TIM? (d) Update the Mode-Path Matrix. (e) Add radiative edge from PCB ( $\varepsilon = 0.9$ ) and re-solve.
13. **Matrix assembly and solution (WE-4.0 extension).** Modify the WE-4.0 network (§04.1.5) by adding a second heat source at Node 2 ( $Q_2 = 3$  W) and a radiative path from Node 2 to Node 4 ( $G_{\text{rad}} = 0.1$  W/K). (a) Re-assemble the  $3 \times 3$  **G** matrix. (b) Solve for the new temperature vector **T**. (c) Has the critical path changed? Compute the new  $\beta$ . (d) Compute  $\Gamma$  at Node 2. Is it still conduction-dominated?

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Chapter 4 of “Thermal Photonics: A Unified Framework for Engineering Heat Flux”  
Book 3 of the Open-Source Engineering Optics Trilogy

Cross-references: Mode-Path Matrix connects every chapter; Eikonal Bridge (Book 1) for optical system analogy; Quantum Field Imaging (Book 2) for fluctuation-dissipation foundation.